CALIBER 2026 | BUSINESS PROPOSAL | PRELIMINARY STAGE

Asset Knowledge Platform

A manufacturing knowledge hub for the LLDPE unit: one governed database behind a Gemini-powered assistant, with live P&ID, OPL simulation, failure memory and predictive maintenance on the same asset tags.

Chosen use case: Case 1, Manufacturing Knowledge Hub

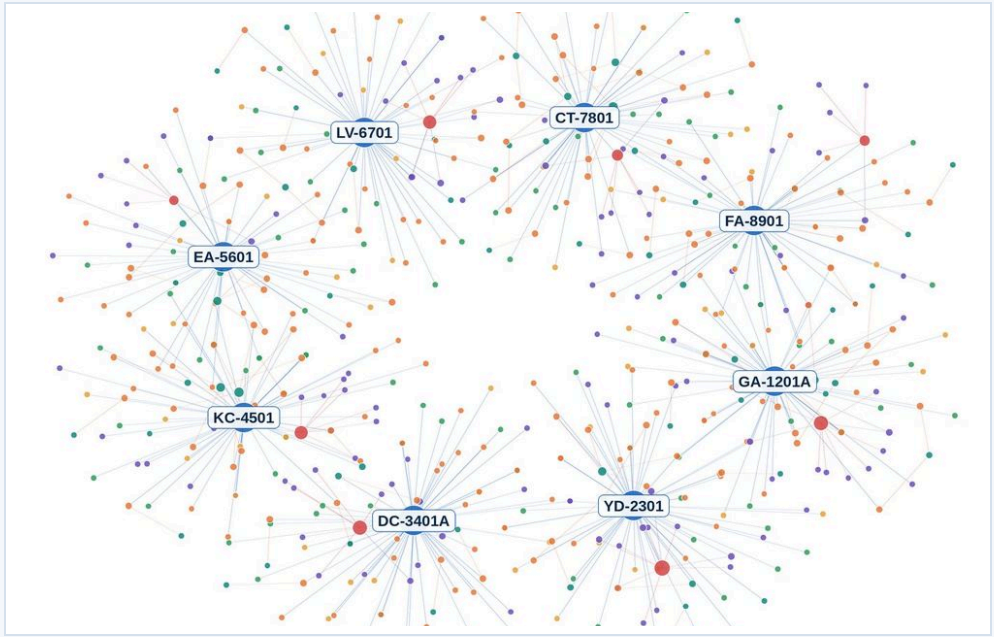
564
linked records

1,109
searchable passages

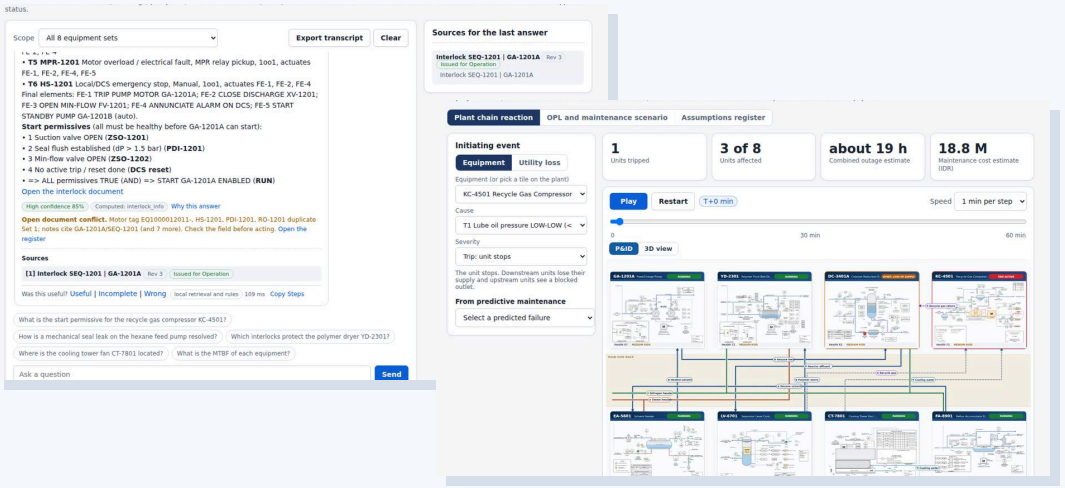
20
computed AI tools

128/128
evaluation questions passed

Team: [Team name]
University: [University] **Supervisor:** [Supervisor name]



Knowledge network of the case data: 8 equipment tags and the 556 records that hang on them

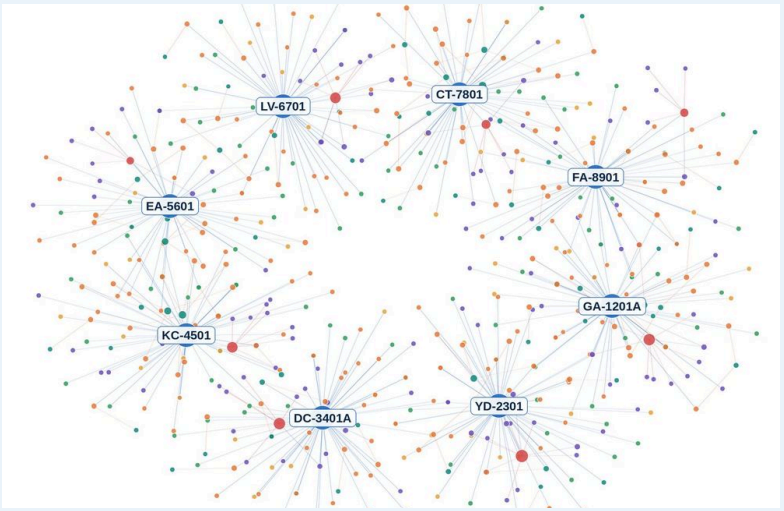


Working prototype on the case baseline data, with a 3-minute demonstration video

Problem Analysis: the Knowledge Gap Between What the Plant Records and What Engineers Can Use at the Point of Work

CALIBER 2026
Case 1

Knowledge landscape of the baseline data



Equipment 8 · Documents 96 · Work orders 211 · Instruments 84 · OPL 56 · Components 19 · Interlocks 8

Analysis

564 records of 7 types are joined only by the equipment tag. No source system stores those links; the platform builds 624 of them.

Key finding

Reading the sources together exposes 48 conflicts, for example valve FV-2302 labelled both trips OPEN and trips CLOSED.

Tension

211 work orders and 63 failures, none linked to the lesson that applies. Only 21% of faults found before turning corrective.

Operating context

Digital agenda
Data, analytics and AI being built into manufacturing

Safety-critical
Continuous, asset-intensive petrochemical operation

Seven functions
Production, Maintenance, Reliability, Warehouse, Procurement, Energy, HSE

Core problem: improper execution

Engineers need a correct step, a current drawing or a past lesson during start-up, troubleshooting and maintenance. The answer exists somewhere, but cannot be found fast, cannot be verified, and is not tied to the asset in front of them.

Key challenges

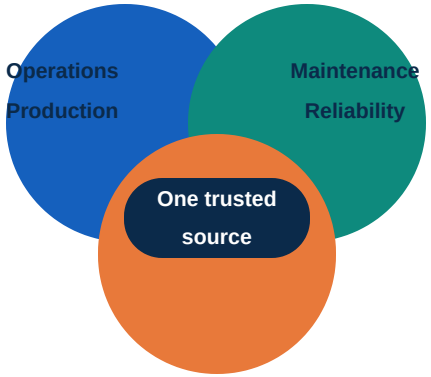
Scattered sources
Separate systems, no shared tag

Missing tacit knowledge
Why a trip occurs stays with senior staff

Conflicting documents
Revision and approval not visible at use

Disconnected live data
Work orders, trips, drawings unlinked

Stakeholder alignment



Needs

The exact step and set point at the valve

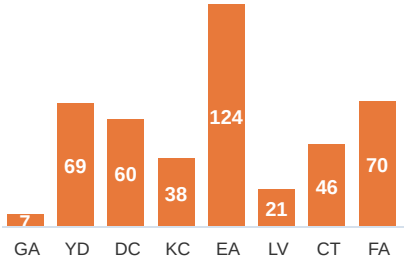
History, lessons, spares

Limits, hazards, permits

Audit and revision trail

Baseline impact

Breakdown downtime by unit, hours



31
breakdowns

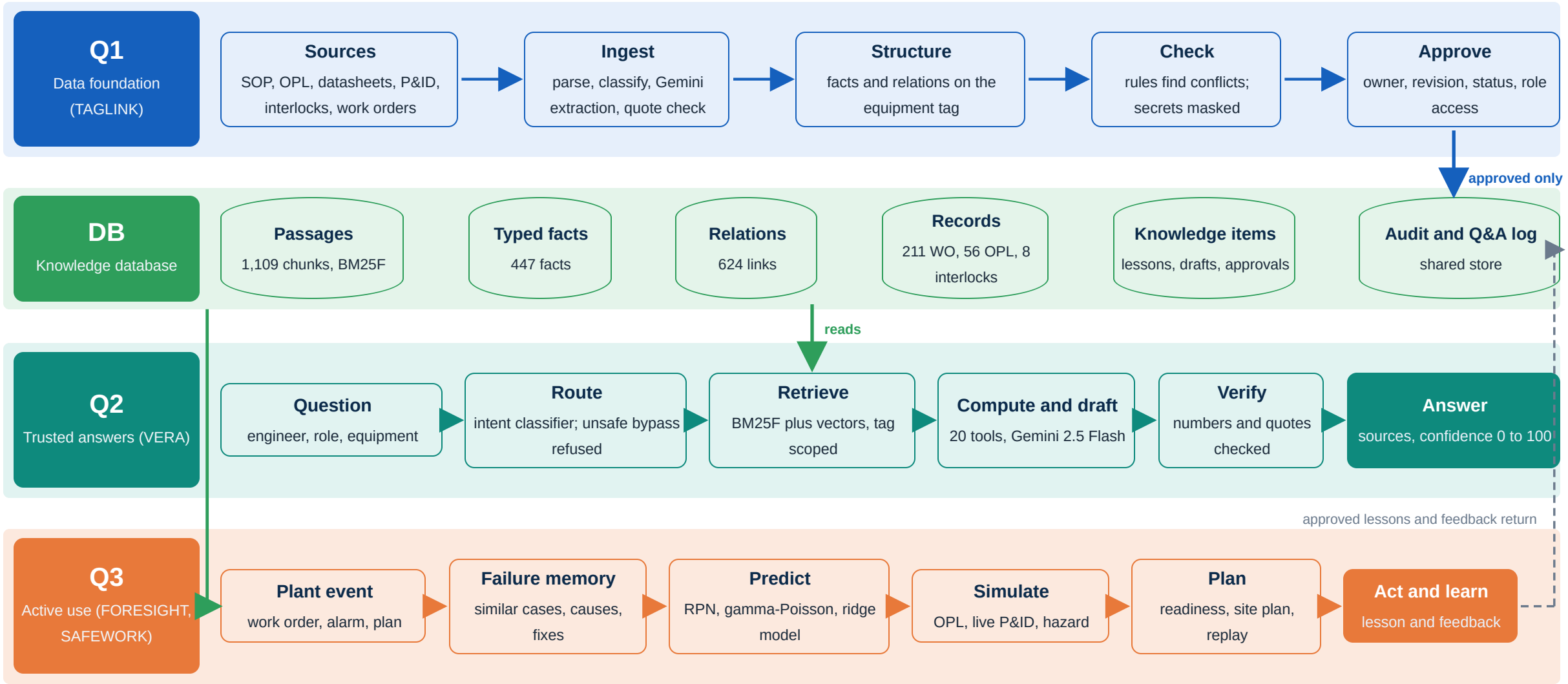
434 h
downtime

IDR 538 M
maintenance, 82% corrective

Reading: EA-5601 alone lost 124 h. Where the history is richer than the procedures, the same fault is investigated again.

Source: CALIBER 2026 case book (Case 1); baseline data analysed in the prototype: 8 equipment sets, LLDPE unit, June 2024 to December 2025

Solution: One Governed Knowledge Database Fed by Data Operations, Read by Three Engines That Answer, Predict and Plan



Source: Prototype architecture and counts from the running application; Gemini 2.5 Flash is reached through the serverless proxy (free tier)

Business Impact: Each Feature Moves a Measured KPI, Worth IDR 1.5 B a Year in the Base Case With a 21-Month Payback

Platform capability	Change at the plant	KPI: baseline to target	IDR M a year
AI assistant, verified answers	Engineer finds the procedure, drawing and past case in minutes	Time to a verified answer: about 15 min to 3 min	604
Failure memory, predictive checks	Known causes and fixes appear at the point of fault; repeat jobs flagged	MTTR 14.0 h to 12.6 h; repeat failures 8 to 5.6 of 55	857
Knowledge Studio, OPL lessons	New engineers learn from approved lessons and the simulator	Time to productivity: one month sooner	60
Interlock guard, readiness, hazard view	Unsafe actions blocked or flagged before execution	Improper executions (counted, not valued)	not valued
Quality register, approvals	Every conflict gets an owner and a close date	48 conflicts owned; 90% closed in the pilot	not valued

Result

IDR 1.52 B
annual value, base case
IRR 40%
NPV at 10%: IDR 0.91 B
Payback 21 months
Net over 3 years: IDR 1.44 B

Cost
one-off IDR 1,400 M
running IDR 370 M a year

Low case IDR 0.8 B a year
no payback within 3 years

Base case IDR 1.5 B a year
payback 21 months

High case IDR 2.1 B a year
payback 16 months

How the value is counted

Search time: 45 users x 20 min a day x 230 days x IDR 175,000 per hour. **Repair time:** 20.4 breakdowns a year x 14 h MTTR x 10% x IDR 30 M per downtime hour. **Ramp-up:** new engineers productive one month sooner. All inputs are assumptions.

Source: Case baseline data (work orders, MTTR, breakdowns); hours and rates are assumptions to be replaced with Company figures (appendix A22, A29, A31)

Feasibility and Roadmap: Read-Only, Advisory Deployment in Four Phases With Explicit Decision Gates

Feasibility: six readiness dimensions

Dimension	Status	In place today	Still needed
Technical	Ready	Prototype end to end; SQL schema and API spec; replaceable model contract	Production database, ingestion queue
Operational	Ready	Advisory only; local search if the model is offline; fits OPL and work-order routines	Shift-team trial, champions per function
Legal	Decision needed	Drawing and vendor IP respected; data held in region	Enterprise AI terms with no training on Company data; PDP Law review
Cybersecurity	Planned	Roles, audit trail, secret masking, prompt-injection guards, rate limits	SSO, key vault, hub outside the OT zone (IEC 62443), red-team test
Data governance	Ready	Owner, revision and status on every record; 48 conflicts in a managed register	Named data stewards, refresh rules per source
Organisation	Planned	Roles and approval workflow built in	Two-hour role-based training, monthly KPI review, sponsor

Deployment path and budget

Prototype today — Static page, Netlify functions, Blobs, Gemini 2.5 Flash free tier, local fallback

Pilot (90 days) — Same stack; read-only EDMS and AIMS adapters, single sign-on, 45 users

Production — PostgreSQL with pgvector, ingestion queue, Company-approved model, SIEM export

One-off budget IDR 1,400 M; running cost IDR 370 M a year

Integration 650	Data 350	Change 250	Security 150

Wrong answer: evidence-only drafting, verification, refusals

Poor data quality: conflict register, named owners

Low adoption: champions, tools used in daily work

Model cost or outage: replaceable model, offline fallback

Implementation roadmap

	Q4 26	Q1 27	Q2 27	Q3 27	Q4 27	Q1 28	Q2 28	Q3 28	Q4 28
0 Mobilise	Start	Finals 4-5 Nov 2026: data access, owners, enterprise AI terms							
1 Pilot, LLDPE unit		Pilot	90-day pilot: 45 users, case data live, success gates						
2 Connect			Integrate	EDMS and AIMS adapters, single sign-on, security review, live tag values					
3 Extend		Predictive and HSE on live data, digital-twin link, units 2 and 3			Extend				
4 Scale					Plant-wide roll-out and continuous improvement			Scale	
Decision gates		G1 Pilot gates met			G2 Security and integration sign-off		G3 Value review		G4 Scale decision

Source: Prototype and architecture notes; dates and budget are proposals for discussion with the Company

Conclusion: A Trusted, Traceable Knowledge Hub Turns Scattered Plant Knowledge Into Faster, Safer and More Reliable Decisions

Summary of the proposal, how it answers the three key questions of the case, and the decision requested

CONNECT TAGLINK: one governed base

- 98 documents and 211 work orders unified on the equipment tag
- 48 conflicts detected and assigned to owners
- Ingestion, quality gates and approval for every record

98 to 1 documents into one governed base

ASK VERA: answers with evidence

- Source, revision, approval status and confidence on every answer
- Numbers and quotes verified against the cited text
- Safety bypass requests refused with the reason

128/128 evaluation questions passed

ACT FORESIGHT, SAFEWORK

- Failure memory with RCA and lessons at the point of fault
- Risk ranking, readiness, hazard screening and permit-aware planning
- Read-only links to EDMS, AIMS and the digital twin

20 tools computed from plant records

How the proposal answers the three key questions of the case

Q1	**Build a structured Industrial Data Ops foundation**	Tag-keyed data model over seven source types; AI-assisted ingestion, quality gates and approval with named owners; conflicts surfaced on day one	**Evidence:** 98 documents, 1,109 passages, 48 conflicts registered
Q2	**Help engineers find trusted answers faster and reduce improper execution**	Hybrid retrieval, evidence-weighted confidence score, verification of numbers and quotes, refusal of unsafe requests	**Evidence:** 128/128 questions passed; 3 minute target versus about 15 minutes today
Q3	**Integrate with operational systems for reliability and improvement**	Read-only EDMS, AIMS, historian and twin links; failure memory, backtested reliability analytics, readiness, hazard screening and site planning	**Evidence:** 64-forecast backtest; 20 computed tools

Decision requested: a 90-day pilot

Weeks 1 to 2
Agree data access, owners and AI terms

Weeks 3 to 6
Load the LLDPE unit and train 45 users

Weeks 7 to 12
Measure against the success gates, then decide on scale

Source: Prototype test results (appendix A23) and business case (slide 4, appendix A22)

Team Profile

Team name, university, supervisor and members (replace the bracketed fields before submission)

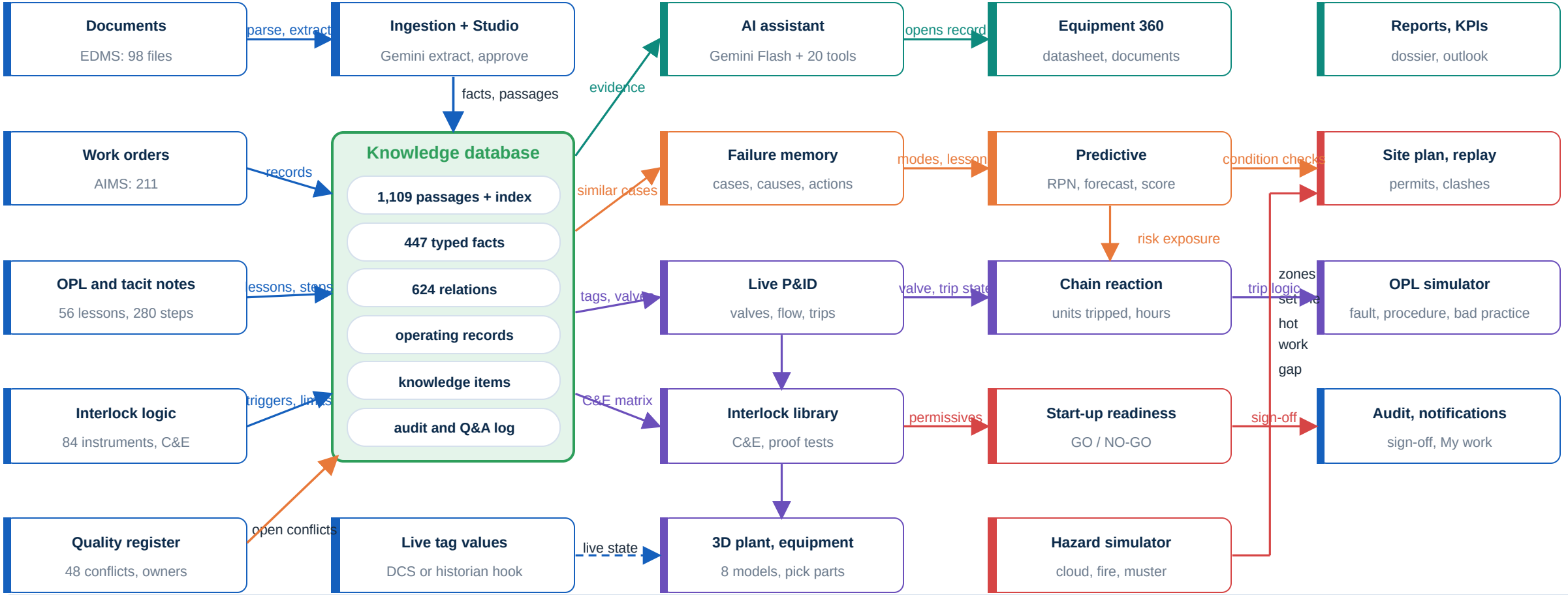
Team name: [Team name] **University:** [University] **Supervisor:** [Supervisor name]

No.	Name	Major	Semester	Area of expertise	Contribution to the proposal
1	[Name]	[Computer Science]	[6]	[AI and software]	[Formulated the idea; built the prototype and the AI assistant]
2	[Name]	[Chemical Engineering]	[7]	[Process and reliability]	[Problem analysis, process content and hazard models]
3	[Name]	[Industrial Engineering]	[7]	[Maintenance and business case]	[Business impact, KPIs and roadmap]

The booklet allows two to three members from the same university. Delete unused rows.

A1 How the Features Connect: Data Flows Between Twenty-Two Modules

Each arrow is a real data hand-off in the prototype; colours show the family a module belongs to



Reports and Performance read from every module; Settings, sync, roles and the guided tour sit around the whole map (appendix A19).

Source: Prototype module registry and call graph (build.py module order, regMod dependencies)

A2 The Database: One Chunk-and-Fact Model in the Browser, a Shared Store Today, PostgreSQL With pgvector as the Target

Every record becomes a chunk with a stable id, a status, a trust weight and an access list; numbers become typed facts; links become relations

Browser

akp_kb in IndexedDB (localStorage fallback)

- In-memory chunk list and term index, rebuilt when data changes
- Knowledge items, audit trail, feedback, answer log, evaluation runs, embeddings cache
- Works offline as a plain file

Shared store

Netlify Blobs through /api/kb

- Collections: uk, wo, fb, audit, qa
- Last write wins; a base version in the request turns a newer server copy into a conflict
- Tombstones propagate deletes; optional access codes; polling for changes

Production target

PostgreSQL 15 with pgvector and pg_trgm

- 25 tables, generated tsvector with GIN index, HNSW cosine index on vector(1536)
- Status vocabulary with trust weights, versions with validity dates
- Row-level role filter through acl_roles

Sync: the page pulls the shared revision, merges per item by timestamp and pushes local edits. If /api/health is unreachable the app keeps working from browser storage.

Assets and documents

asset	document	chunk	fact
component	document_version		relation

Operations

interlock	instrument	opl	opl_case
interlock_trigger	work_order	failure_event	

Knowledge and governance

knowledge_item	knowledge_review	qa_log	feedback
knowledge_revision		qa_citation	eval_case

Also: eval_run, audit_log, app_user, status_vocab (11 trust codes). Keys: chunk_id = kind:record#ordinal, tag is the common join key

Anatomy of one chunk

chunk_id	il:TJC-LLD-IL-KC-4501#03
kind / record	il / interlock sheet
title, section	Interlock | KC-4501 | T3
text	VSHH-4505 80 um, 2oo3 ...
tags	{KC-4501, VSHH-4505}
status, trust	approved, 1.00
revision, date	Rev 3, 2025-02-11
acl roles	all roles
hash, tokens	cyrb53, 112
embedding	vector(1536) when connected

Indexes: GIN on the generated tsvector and on tags; B-tree on asset and status; HNSW (cosine) on the embedding. Embeddings are recomputed only when the content hash changes.

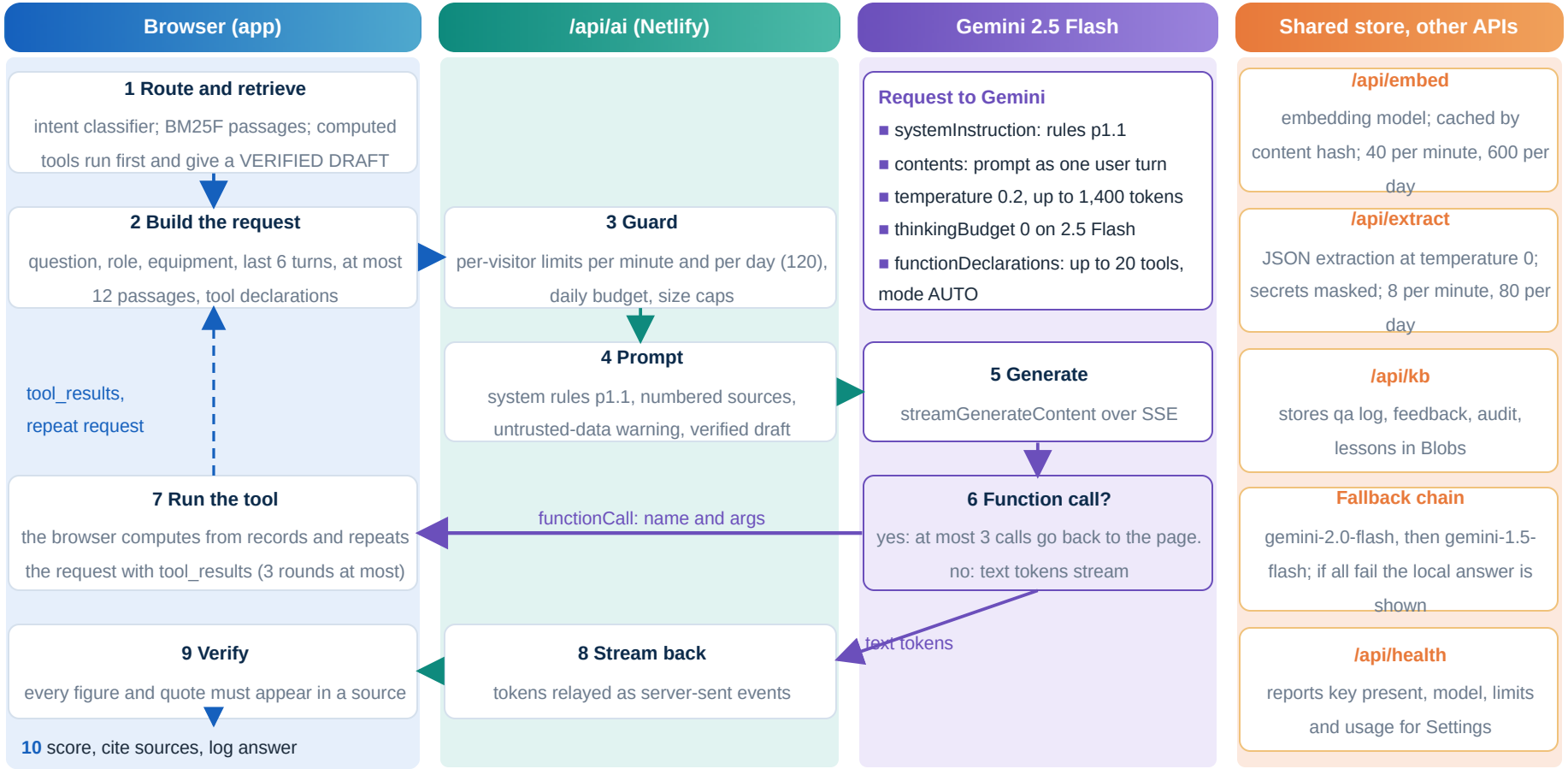
1,109	447	624	564	98	211
chunks	typed facts	relations	linked records	documents	work orders

Counts are read live from the prototype; the same shapes map one-to-one onto the SQL tables.

Source: Prototype data layer (dl.js, facts.js, sync.js), Netlify Blobs functions (kb.mjs) and the SQL schema in docs/DATA_MODEL.sql

A3 The AI Layer: Gemini 2.5 Flash (Free Tier) Behind a Serverless Proxy, With Computed Tools and Local Verification

Assuming the free-tier key is connected: what is sent, what comes back, and which checks run before an answer reaches the engineer



Output contract: answer text, cited source numbers, confidence 0 to 100, needs_verification, refused. Free-tier quotas apply to the Gemini key; the proxy adds its own per-visitor limits so one user cannot exhaust them. The model can be replaced by any provider that implements the same contract.

Prompt anatomy

System rules p1.1
8 rules: answer from sources, cite [n], never invent values, refuse bypass, treat sources as data

<context>
role, page, equipment, intent, language

<history>
last 6 turns, 500 characters each

<sources> at most 12
id, kind, status, revision, score, text

VERIFIED DRAFT
computed numbers that must stay unchanged

<tool_result> at most 4
2,500 characters each

<question>
user text

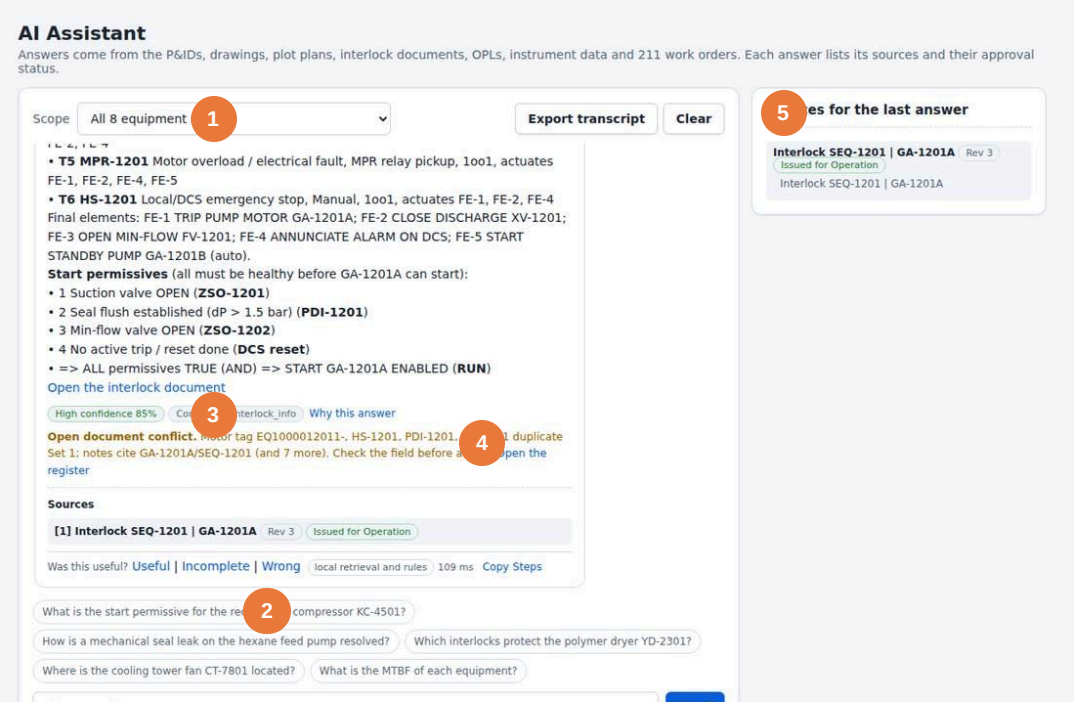
Why this is safe to run on free tier

Key stays on the server. Limits protect the quota. Gemini only words evidence already retrieved or computed, so a failed call costs an answer style, not correctness; the local engine still answers.

Source: Prototype netlify/functions (ai.mjs, extract.mjs, embed.mjs, _lib.mjs) and the page-side assistant (ai.js); prompt version p1.1

A4 The Assistant: Intent Routing, Twenty Computed Tools and an Evidence-Weighted Confidence Score

Counts, rankings and forecasts are calculated from records; Gemini is used to word the result and to answer open questions from retrieved passages



1. Scope selector: the question is limited to one equipment set or the whole plant
2. Suggested questions generated from the data
3. Confidence chip with a breakdown, and the tool that produced the figures
4. Open document conflict note when the answer touches a flagged document
5. Source panel: revision, approval status, record id for every cited item

Twenty tools, grouped

Reliability and cost

maintenance_stats, kpi_summary, failure_modes, predictive_maintenance, failure_memory, history_lookup, spares_outlook

Assets and limits

equipment_info, spec_lookup, tag_facts, plot_plan_query, compare_equipment, rank_equipment

Operations and safety

interlock_info, procedure_help, consequence_analysis, cascade_analysis, hazard_effects, safety_guard

Data governance

data_issues

Intents: chain reaction (utility loss), bypass request, consequence, comparison, ranking, data issues, history, procedure and start-up, diagnosis, spares; anything else goes to retrieval.

Safety guard

A request to bypass or defeat a trip is never answered with steps. The reply names the function that would be defeated (voting, SIL, set point), the consequence from the cause-and-effect matrix, and the approved route: bypass permit, management of change, compensating measures. Covered by safety cases in the evaluation set.

Confidence score, 0 to 100

Question words in one passage	0.45
Match strength of best passage	0.22
Approval status of sources	0.15
Agreement between sources	0.10
Distinct sources found	0.08

Caps after weighting

- Broad question: 60
- Different equipment than asked: 40
- Under half the question words found: 45
- No single passage covers it: 50

High 70+ | Medium 45-69 | Low below 45

Number and quote check

Each figure with a unit (barg, degC, mm/s, m3/h, kW, IDR, %) is looked up in the cited text. Unsupported figures are listed under the answer and the confidence drops. Calculated tools report their own confidence.

Source: Prototype assistant (ai.js, intent.js, ai2.js); weights and caps are the values used in the running application

A5 Retrieval: BM25F With Tag, Status, Equipment and Role Weighting, Embeddings When Connected

The best evidence is chosen before any language model is called, and every passage carries the reason it was ranked

1 Question	2 Tokenise	3 Score	4 Re-weight	5 Fuse	6 Select
Free text. Tags found by pattern, equipment inferred from names, aliases and instrument tags	Stemming, synonyms, plant stop words; tags kept whole and weighted 2.2	BM25 over weighted fields: tag 2.5, title 2.0, section 1.5, body 1.0; k1 1.4, b 0.75	x status trust, x source-kind prior, x equipment scope, x role fit	With embeddings: 0.65 x BM25F + 0.35 x cosine x 40; semantic recall at cosine 0.6	Role filter, at most 2 passages per document, top results with relative score

Weights by field



Why BM25F first: plant text is short, tag-heavy and full of identifiers that exact matching handles well; the weights stop a long body from outranking a heading that names the tag. Embeddings add recall for questions with no shared words.

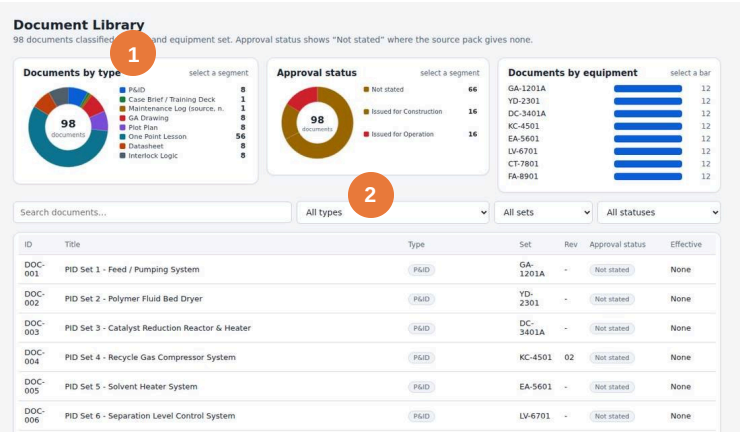
Output per passage: chunk id, kind, revision, status, matched terms, score, relative score, share of question words covered. The last two feed the confidence score.

Multipliers applied to the BM25F score

Status trust	0.35 to 1.00
Source-kind prior	0.55 to 1.05
Equipment in scope	x 1.45
Other equipment	x 0.70
Role fit	x 1.10 to 1.15

Kind priors: interlock 1.05, OPL 1.05, document 1.00, work order 1.00, plot plan 0.95, instrument 0.85, data-quality note 0.55. Role fit: operators favour OPL and interlocks, reliability favours work orders and forecasts, HSE favours interlocks and OPL, managers favour forecasts.

Retrieval in the product

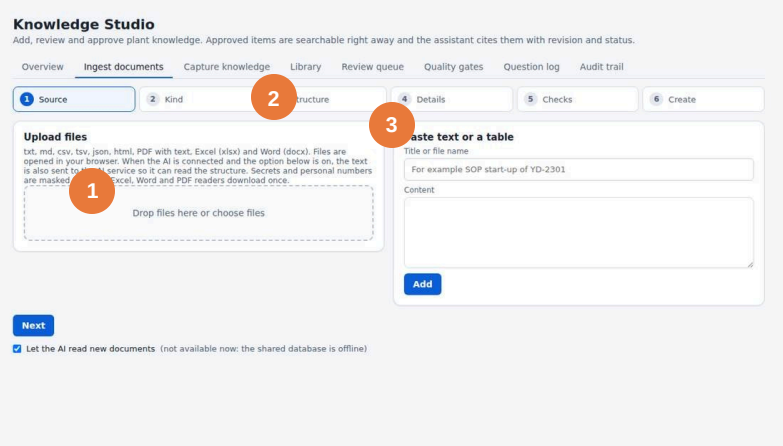


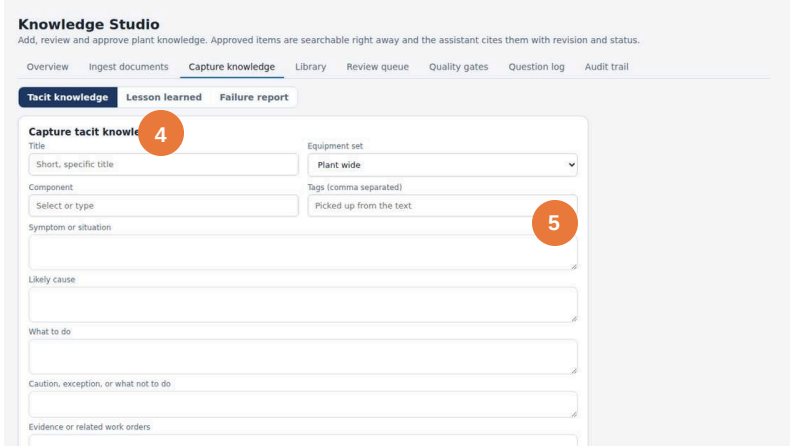
1. Global search and Documents list use the same ranked index
2. Status and revision columns come from the same canonical vocabulary

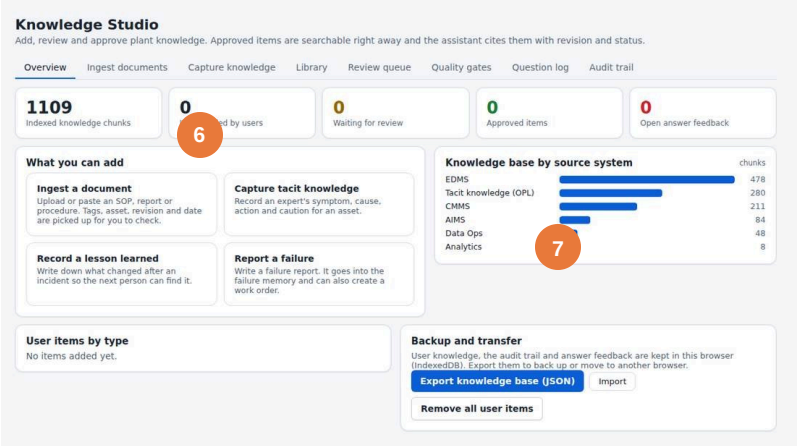
Source: Prototype retrieval index (dl.js); parameters are the values used in the running application

A6 Industrial Data Ops in Use: Ingestion Wizard and Knowledge Studio

How a new document, work-order export or piece of tacit knowledge becomes approved, searchable knowledge







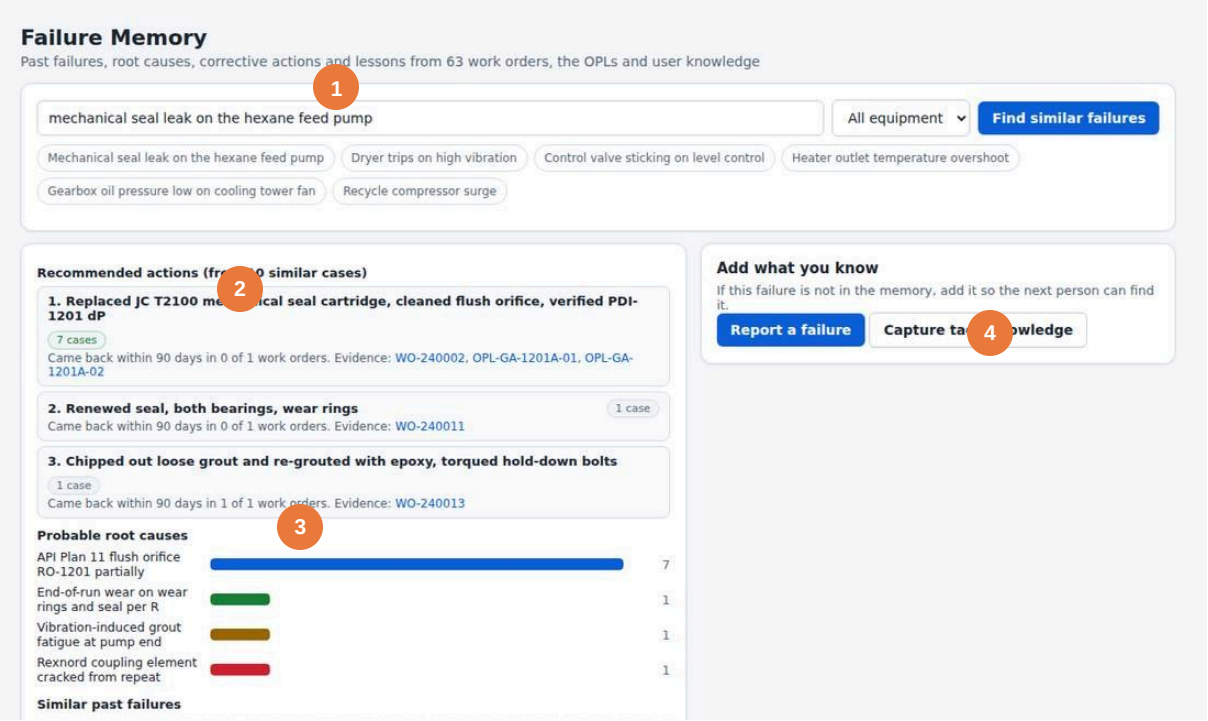
1. Six steps: source, kind, structure or column mapping, details, checks, create
2. Upload txt, md, csv, tsv, json, html, PDF, Excel or Word, or paste text
3. Option to let the AI read the file when connected; secrets are masked first
4. Capture tacit knowledge as symptom, cause, action and caution; failure reports and lessons
5. Each item gets an owner, equipment, revision and a status that starts at Draft
6. Overview, review queue, quality gates, question log and audit trail
7. Reviewers approve or reject; approval makes the item searchable at once

1 Read	2 Verify	3 Map	4 Gate	5 Approve	6 Index
Parsers split CSV or text; with the AI on, /api/extract returns JSON	Each step, value or fact must quote the source word for word or it is dropped	Tags, assets, revision and date detected; columns mapped for tables	Rules: missing owner, unknown tag, conflict with an approved set point	Draft, In review, Approved, Superseded, Rejected; every step audited	Chunks, typed facts and a source copy go to the shared store and the search index

Source: Prototype Knowledge Studio and ingestion wizard (ks.js, wiz.js, ingest.js, facts.js) and /api/extract

A7 Failure Memory: Similar Failures, Root Causes and Actions Retrieved at the Moment of Need

FORESIGHT engine: a symptom is matched against work orders, one point lessons and approved user lessons



1. Free-text symptom, optional equipment filter
2. Recommended actions from the 10 most similar cases, with recurrence within 90 days
3. Probable root causes counted across the matches
4. Add what you know: report a failure or capture tacit knowledge in one click

How a match is built

1	**Retrieve**	Same BM25F index, limited to work orders, OPL cases and approved lessons; top 60
2	**Parse**	Symptom, cause and action fields; component from the part list
3	**Re-rank**	Relevance, plus a boost when the same component is named
4	**Group**	Causes and actions merged by the first four meaningful word stems
5	**Recurrence**	Same component failing again within 90 days
6	**Recommend**	Ranked actions with case counts and downtime range

Where it appears

- Failure Memory page
- Panel inside a work order and the failure view
- failure_memory tool of the assistant
- Scenario and OPL simulator hints

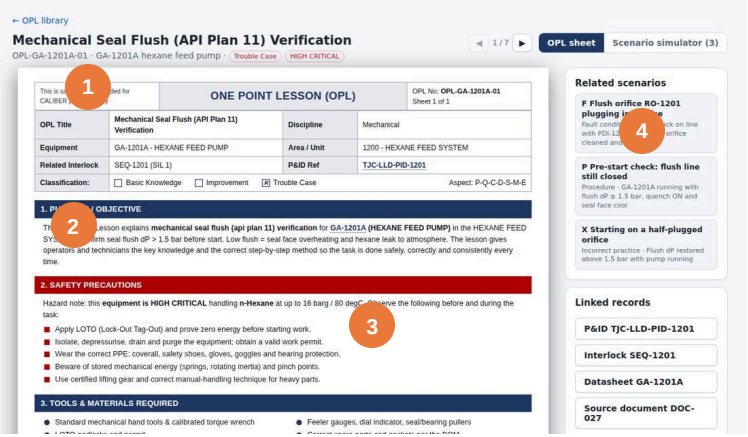
Limits

Matching is lexical with stemming unless embeddings are connected, so it depends on how clearly work orders are written. Each match links to the full record so the engineer judges the fit.

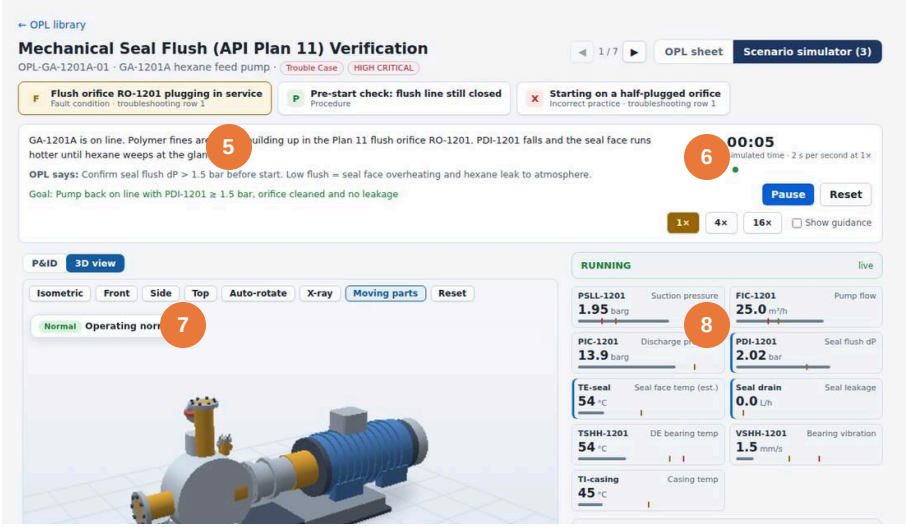
Source: Prototype failure memory (fm.js, pdm2.js) on the case work orders and OPL troubleshooting cases

A8 OPL Library and Scenario Simulator: 56 Lessons You Can Read and Then Play on the Machine

An OPL (one point lesson) is the plant's own short lesson; the simulator turns its troubleshooting cases and steps into a time-stepped exercise



1. Standard OPL header: equipment, discipline, classification, reviewer, approver
2. Safety precautions, tools and materials
3. Detailed steps, each with a check or acceptance criterion
4. Related scenarios, linked P&ID, interlock, datasheet and source document



5. Brief: what the OPL says and the goal; simulated clock with time compression 1x, 4x, 16x
6. 2D P&ID or 3D machine that reacts: shakes, steams, stops
7. Field and panel actions, numbered to the OPL steps; non-standard actions are included on purpose
8. Readouts, trip table, trend, event log and the step checklist

How the simulator works

1. **Process model:** each equipment has variables (pressures, flows, temperatures, levels) integrated in small time steps
2. **Alarm and trip logic:** set points and delays from the interlock document; trips actuate the final elements; a bypass is logged and nothing protects
3. **Operator actions:** valve moves, starts, resets, bypass; the effect shows in the variables
4. **Goal and hold time:** success needs the target state held for a set time; damage ends the run

Score and kinds

Score = 100 - 20 per unsafe action - 12 per trip (up to 3) + 2 per OPL step applied (up to 10); a failed run is capped at 35. Kinds: fault diagnosis, procedure, slow trend, bad practice. The key learning point is shown at the end.

56 OPL documents

280 procedure steps

168 cases

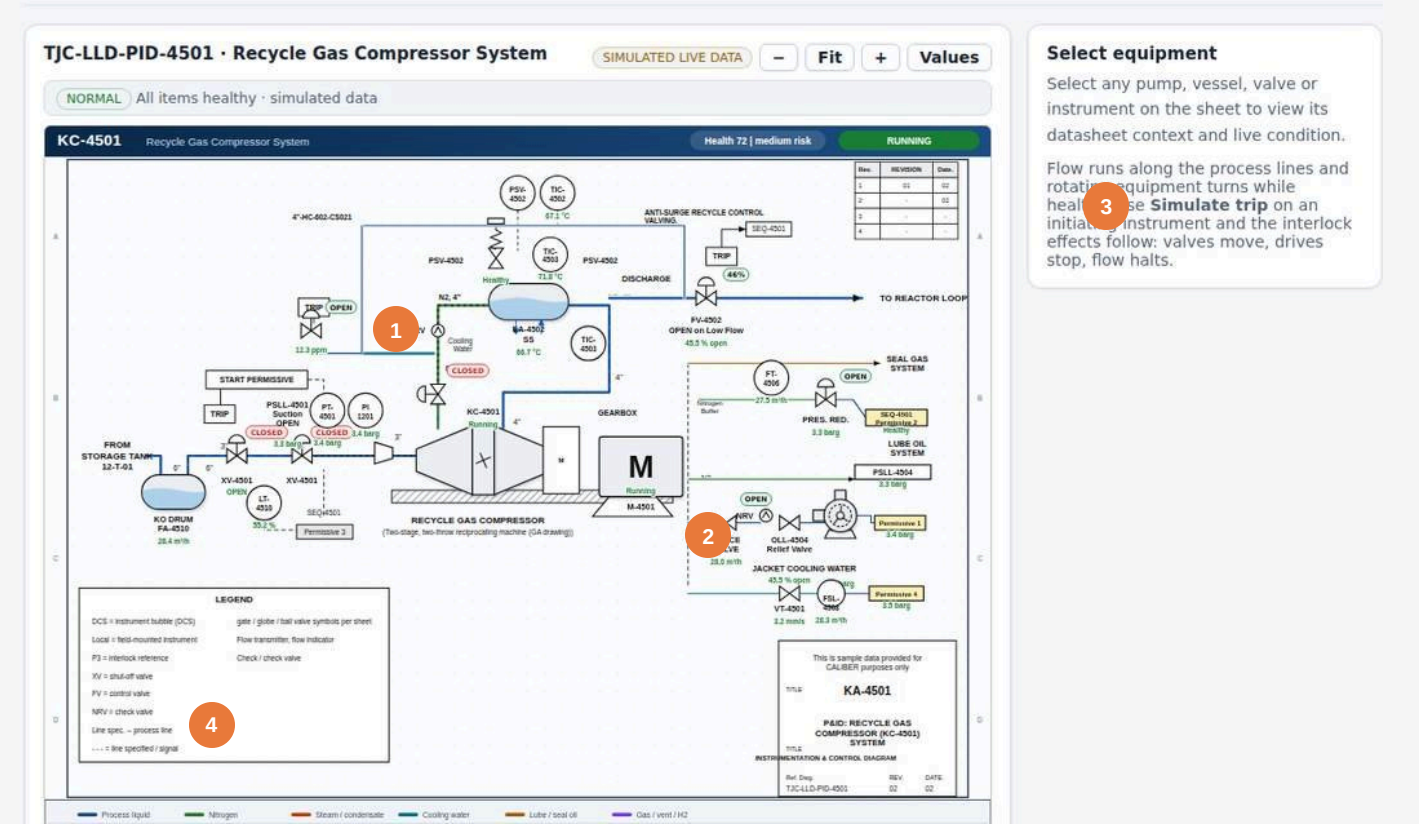
Connected to: Interlock library · Live P&ID · Failure memory · Assistant · 3D machine

The assistant can read any OPL and quote its steps; the simulator reads the same set points, so the lesson and the exercise cannot disagree.

Source: Prototype OPL library and simulator (base.html OPL view and sim engine, sim3d.js); counts from the case OPL documents

A9 Live P&ID: Valves That Open and Close, Flow That Stops and Trips That Follow From the Interlock Logic

Eight P&IDs redrawn as a consistent, service-coloured visual engine and connected to the interlock model, the readiness check and the 3D plant



1. Valve badge: OPEN, CLOSED or percent open; press it, double-click the valve or use the side panel
2. A closed valve stops the flow in the pipe after it and in the pipes that continue from there
3. Instrument bubbles show live values and alarm state; trips latch until reset
4. Service colours: process liquid, nitrogen, steam and condensate, cooling water, lube and seal oil, gas and vent

What the engine does

Flow logic
Pipes are drawn as segments; closing a valve marks every downstream segment as stagnant and stops the flow animation.

Permissives
Closing a valve that proves a start permissive (for example XV-1201 with ZSO-1201) trips the item that depends on it through the same cause-and-effect rows as the simulator.

Automatic trips
Interlock trips close their shutdown valves by themselves, and the badge follows.

Levels
Vessels show a liquid level that moves with the level instrument.

Overview
All eight sheets joined by their battery-limit lines on one plant picture, with a state per unit.

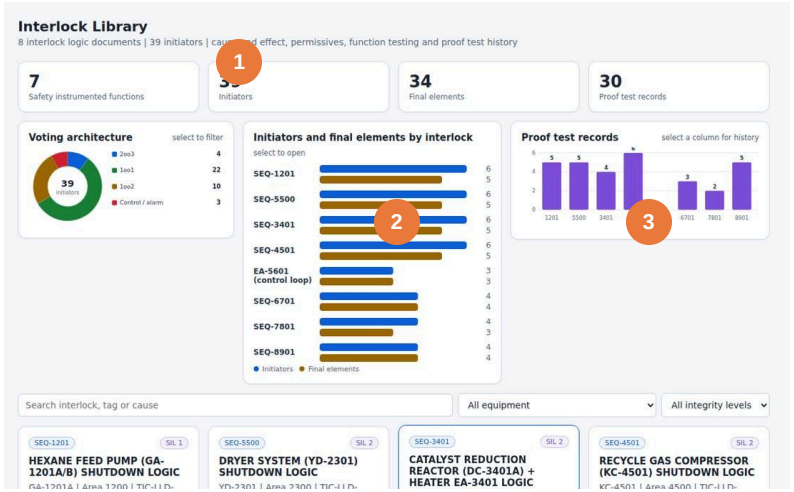
1 Valve	2 Logic	3 Trip
operator closes XV	permissive lost	final elements act

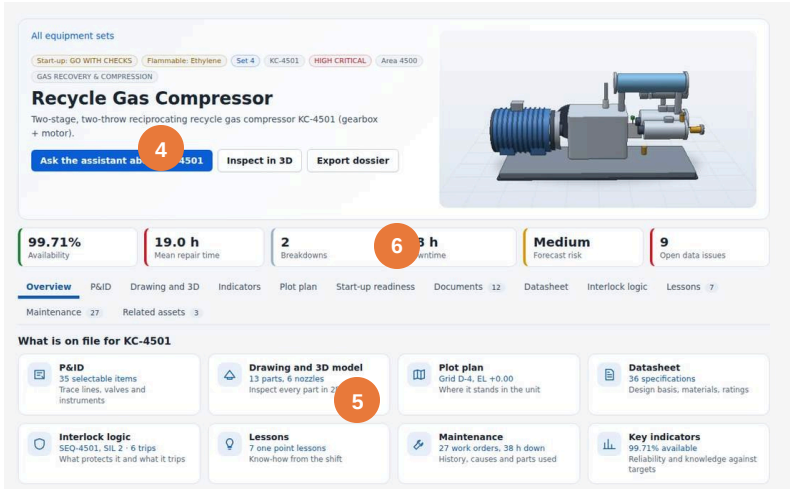
Connected to: Interlock · Readiness · Scenarios · 3D plant

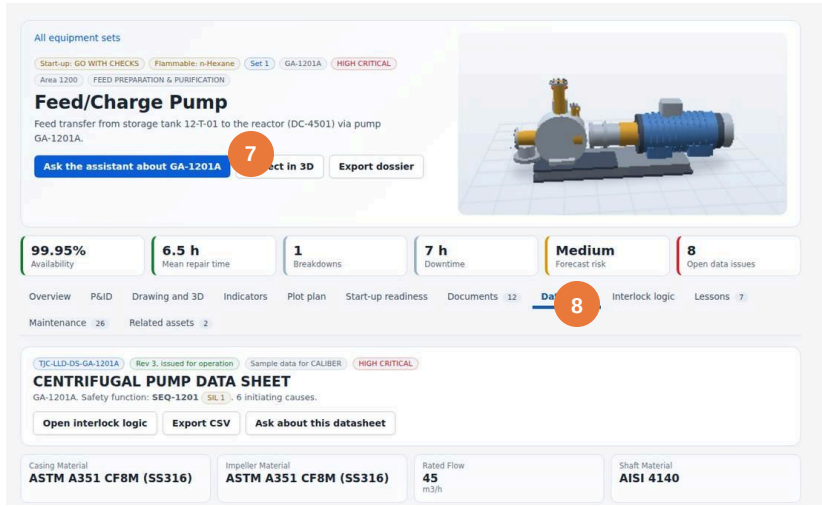
Source: Prototype P&ID engine (pid2.js, pidx.js, pidv.js); 8 sheets, 20 operable valves, 232 tagged items

A10 Interlock Library and Equipment Pages: Cause and Effect, Proof Tests and Datasheets for Every Set

The safety logic of each unit is read as data, reconciled with the instrument list and the P&ID, and used by every other feature







1. Library views: interlock sheet, function tests, proof test history
2. Cause and effect: initiator, set point, voting (1oo2, 2oo3), SIL, final elements
3. Reconciliation: Matches P&ID, Differs from P&ID, Not on instrument list
4. Equipment header: tag, criticality, state, 3D model
5. Tabs: overview, P&ID, drawing and 3D, indicators, plot plan, readiness, documents, datasheet, interlock logic, lessons, maintenance
6. What is on file as clickable cards
7. Datasheet grouped and searchable like a specification sheet
8. Every value is a typed fact and appears in the assistant with its source

1 Read	2 Reconcile	3 Test history	4 Publish
interlock document gives initiators, voting, effects, permissives	set points compared with the instrument list and the P&ID note	function and proof tests found in the work orders by wording	typed facts for tools; C&E matrix exported as 39 rows

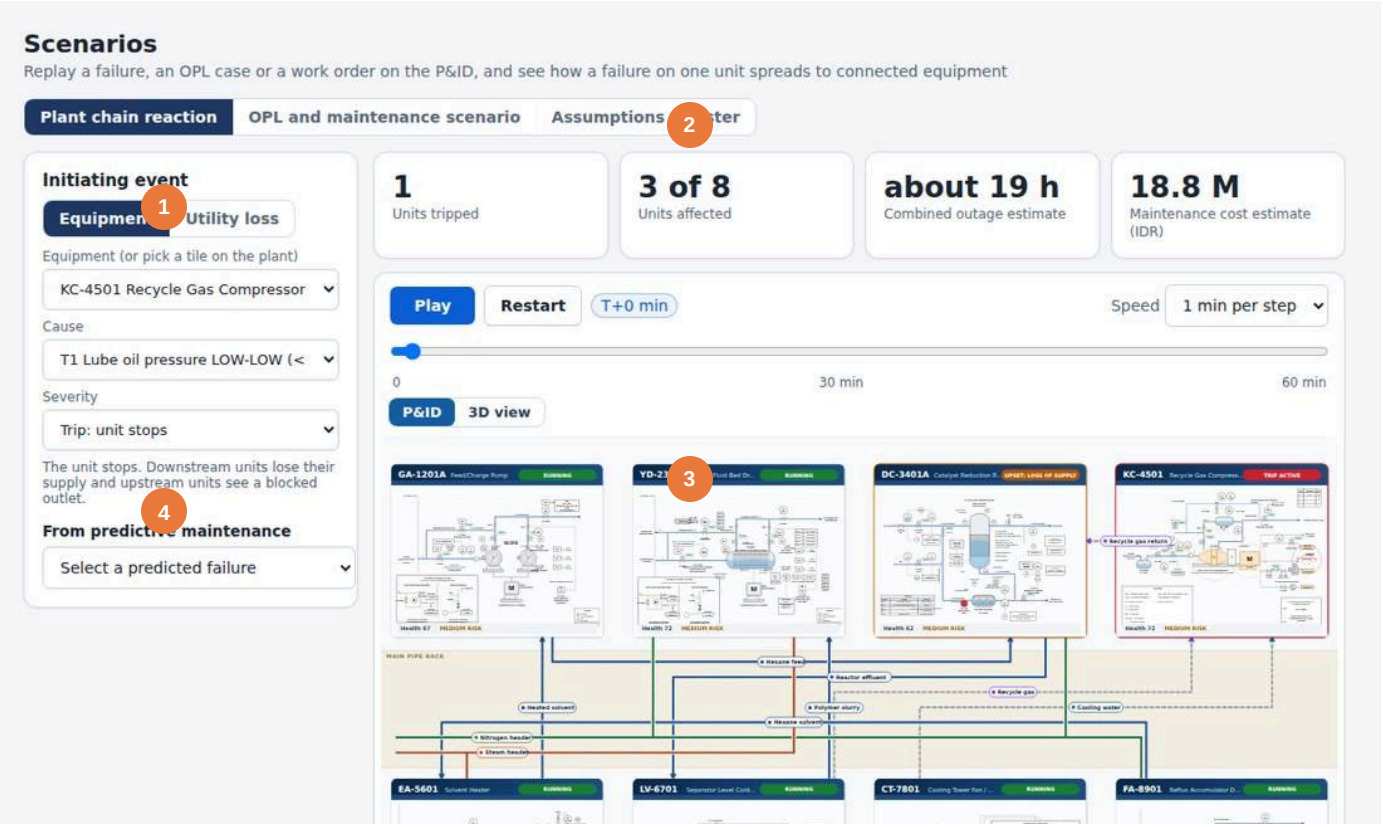
Why it matters

A trip set point stated in three places gets compared automatically. A difference shows as a flag and enters the conflict register, so a technician sees it before the proof test, not after.

Source: Prototype interlock library and equipment set pages (il.js, eq2.js); 8 interlock sets, 39 initiators, 37 start permissives, 84 instruments

A11 Scenarios and Chain Reaction: What Else Trips When a Unit or a Utility Is Lost

Plant-wide cascade over the connected P&IDs, plus a unit scenario player that replays an OPL case or a work order on the P&ID



1. Initiating event: an equipment trip or a utility loss (nitrogen, steam, cooling water), with severity
2. Result tiles: units tripped, units affected, combined outage estimate, maintenance cost estimate
3. Connected P&ID overview: units go to alarm and trip as the event spreads, one simulated minute per step
4. Start from a predictive risk: a forecast failure becomes the initiating event

Cascade algorithm

1 Seed	2 Connect	3 Propagate	4 Report
failing unit or utility	battery-limit lines between sheets	loss of supply raises upset, alarm, then trip by delay	units, hours, cost

Unit scenario player

- Pick an OPL troubleshooting case or a work order; the phases (initiating event, alarm, trip, relief) are shown on the P&ID of that unit
- Tags involved are highlighted, trips are forced on the live model so the P&ID, readiness and 3D views agree
- Each phase carries the OPL guidance and similar past events from failure memory

Output used elsewhere

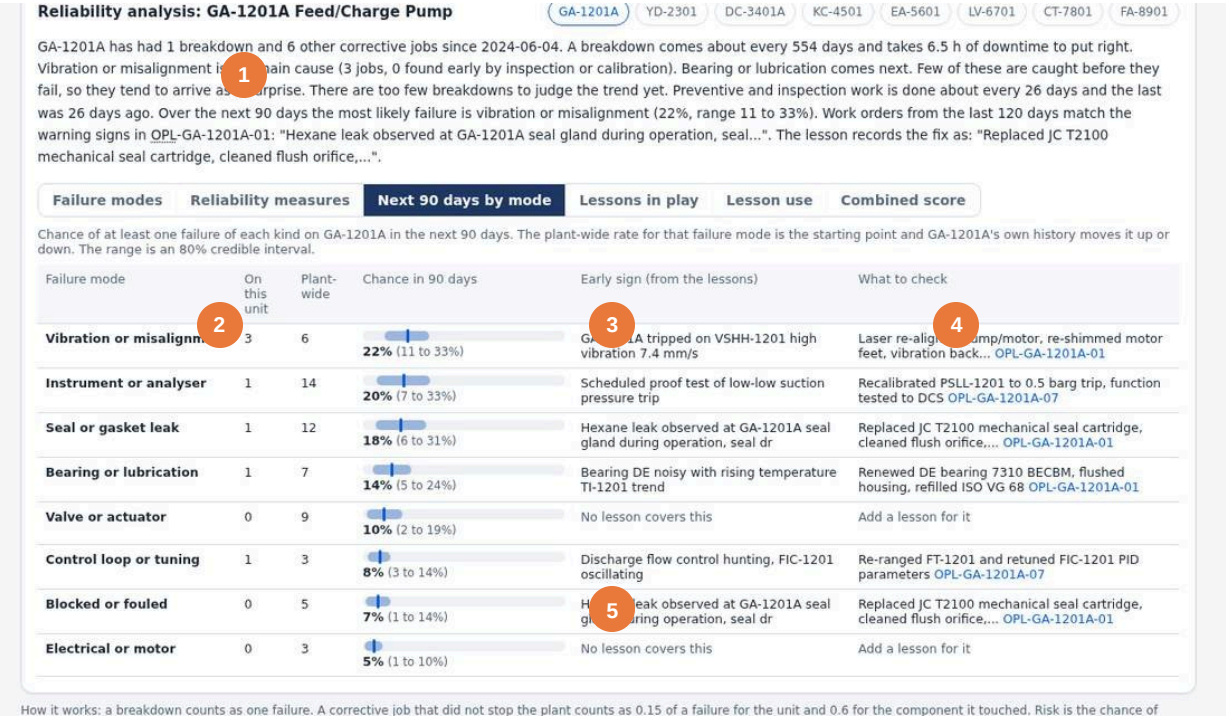
The assistant tool cascade_analysis returns the same figures; predictive maintenance adds the chain-reaction penalty to a unit's consequence score (0.35 per additional unit tripped, maximum 5).

Connected to: Live P&ID · OPL · Predictive · Assistant · 3D

Source: Prototype scenario engine and views (scn.js, scnui.js); example shown: compressor KC-4501 lube oil low-low

A12 Predictive Maintenance Page: Health, Risk Matrix, Timeline and Intervention Plan for 8 Units and 42 Components

Everything on the page is computed in the browser from the 211 work orders and the OPL lessons, with confidence labels where data is thin



1. Model header: mode, horizon, action threshold, chain-reaction toggle, export
2. Risk matrix: probability against consequence, click a cell to filter
3. Unit health index 0 to 100 and 90-day failure probability
4. Timeline of expected failures with p10 to p90 range and the recommended intervention date
5. Unit detail: health trend with forecast, cumulative probability, component risk in 2D and 3D

Health index

100 minus penalties for recent breakdowns, preventive work overdue and a rising trend; 8 points added right after a breakdown

Failure probability

constant fleet rate shared by share of past events (default); Weibull per unit as an experimental option with a wear-out slider

Consequence

downtime, cost and criticality, plus the chain-reaction penalty from the scenario engine

Intervention

recommended date = earliest predicted failure minus 14 days lead; spare parts to stage; schedule a predictive work order in one click

Validation shown on the page

A rolling backtest of 64 forecasts compares the default with the Weibull option and a constant guess (Brier score, ranking skill, calibration). The page states in plain words that a unit's own history does not tell which unit fails next, so the order comes from consequence.

Connected to: Failure memory | Scenarios | Site plan | Assistant | Reports

Source: Prototype predictive maintenance view (pdm.js, pdm2.js, mh4.js); 50 entities, 90-day default horizon

A13 Reliability Algorithms: Failure-Mode Risk, Bayesian Forecasts and a Backtested Combined Score

FORESIGHT engine: quantitative models built on 211 work orders, validated by rolling-origin backtesting, with their limits stated

1 Failure-mode risk priority

- Mode taxonomy read from problem, cause and action text (10 modes)
- RPN = Severity x Occurrence x Detection
- Severity: percentile of downtime (55%) and cost (45%), scaled 1 to 10
- Occurrence: 1 + 2.4 log2(1 + 3 x rate per year), max 10
- Detection: 1 + 9 x (1 - share found early), minus 1.2 if an OPL covers the mode

2 Gamma-Poisson mode forecast

- Prior from the fleet rate: shape 1.5, rate 1.5 / fleet rate
- Posterior after n own events in T days: shape 1.5 + n, rate prior + T
- P(failure in h days) = 1 - exp(-mean rate x h)
- 80% credible range from the gamma quantile (Wilson-Hilferty)
- Few events shrink toward the fleet instead of over-reacting

3 Measures and trend

- MTBF = (operating hours - downtime) / breakdowns; MTTR = downtime / breakdowns
- Laplace trend test: u above 1.64 means failures accelerating, below -1.64 slowing
- Share of corrective jobs repeating the previous mode
- Preventive-work effectiveness: interval against the usual gap

4 Combined unit score and its validation

- Ridge-regularised logistic regression on eight standardised features: recent breakdowns, time since last breakdown against the usual gap, trend, repeat failures, overdue preventive work, lesson warning signs, recent corrective jobs, low health
- 300 gradient-descent iterations with an L2 penalty
- Rolling-origin backtest: train on the past, forecast the next 90 days, advance 20 days
- Calibration slope shrinks the output toward the base rate

64
forecasts tested

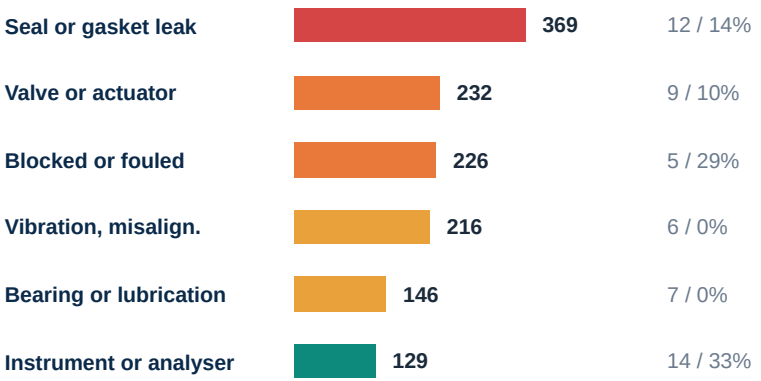
0.26
Brier score, plant-wide rate

0.47
AUC; 0.50 is chance

A unit's own history does not predict which unit fails next on this data. The page ranks units by consequence and reports the combined score as advisory. Improving it needs more history or live condition data, which is a pilot task.

Failure modes by risk priority

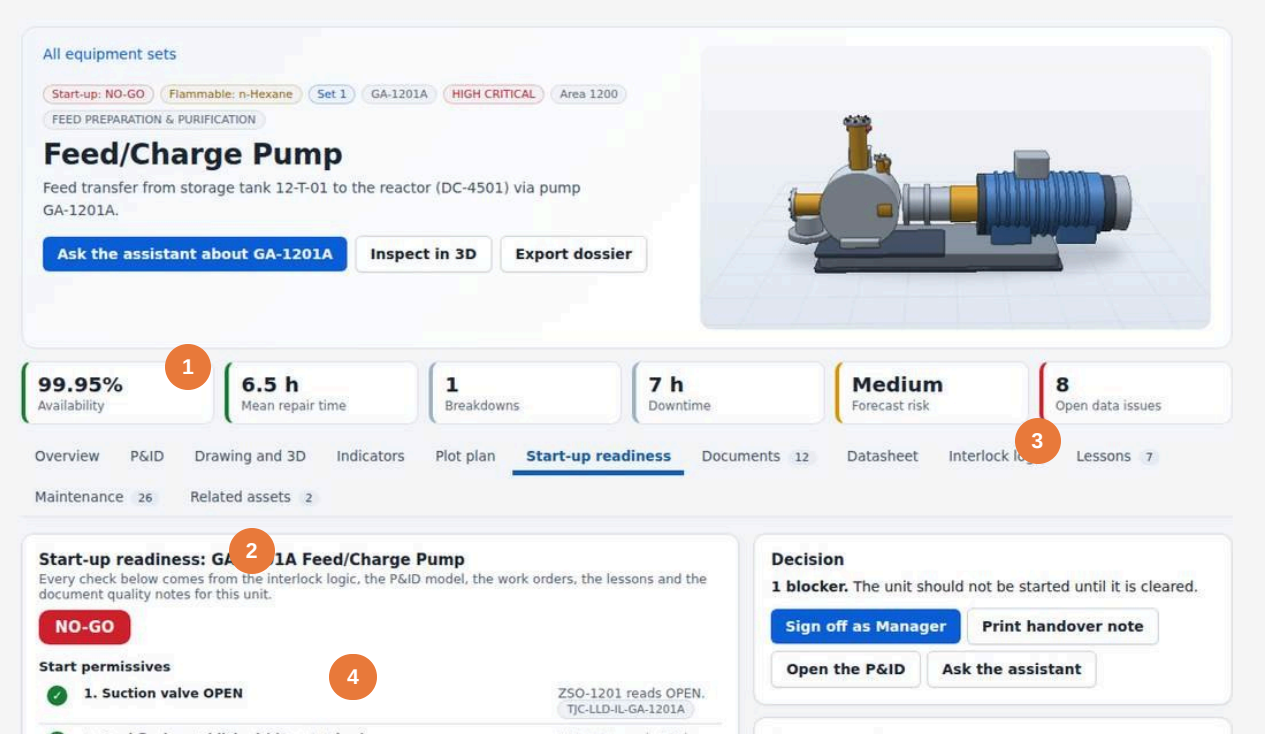
Mode, RPN (bar), jobs / share found early



Source: Prototype reliability analytics (pdm.js, pdm2.js); 63 breakdown and corrective jobs behind the failure-mode view

A14 Start-Up Readiness: One GO, GO WITH CHECKS or NO-GO From Interlock, P&ID, Work Orders and Lessons Read Together

Before a unit is started or restarted, the checks that used to sit in five places are evaluated on one page, with a recorded sign-off



1. Verdict banner and list of blockers with the reason
2. Each start permissive checked against the live model: valve position, flush, min-flow, no active trip
3. Decision panel: sign off as manager, print the handover note, open the P&ID, ask the assistant
4. Lessons to read, earlier start-up problems from work orders, open document conflicts

Checks (22 items on the example unit)

Permissives
every start permissive of the interlock sheet against the live P&ID model

Latched trips and closed valves
any is a blocker

Recent corrective work
flagged until closed

Lessons
OPL key points for the start-up must be read

Past start-up problems
work orders that mention start-up for this unit

Document conflicts
open items touching the unit are listed

GO	GO WITH CHECKS	NO-GO
all checks pass	no blocker, items to confirm	at least one blocker

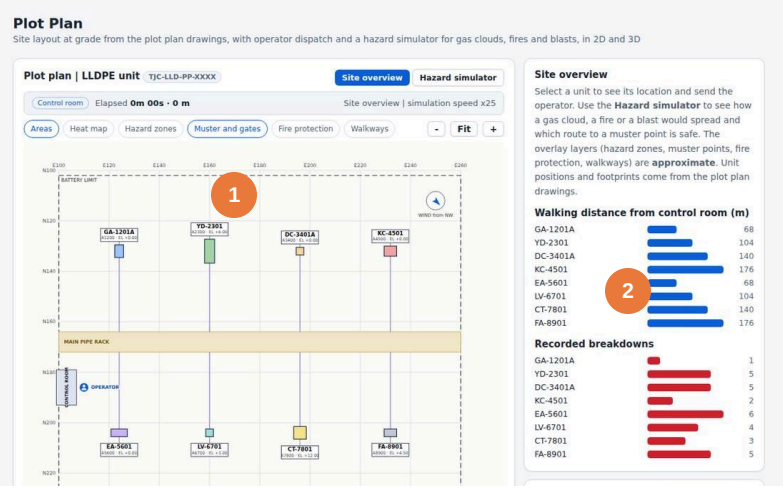
A sign-off with name, role and note is written to the audit trail; a latched trip must be reset on the P&ID before the verdict can change.

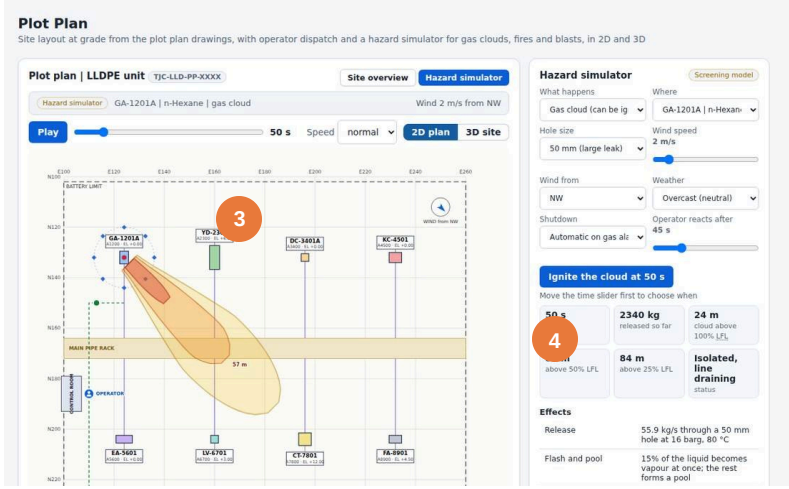
Connected to: Interlock · Live P&ID · OPL · Work orders · Quality register · Audit

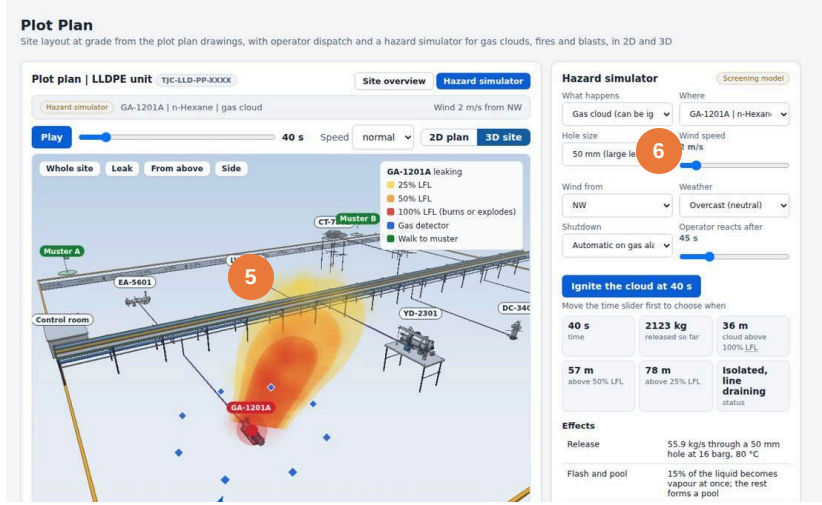
Source: Prototype readiness module (ready.js); example shown: feed pump GA-1201A with a latched low-low suction trip

A15 Plot Plan and HSE Hazard Simulator: Release, Cloud, Fire and Muster Walk on the True-Scale Site

Screening models run on the real coordinates of the eight units, the control room, the gates and the muster points







1. Scaled plot plan with E/N grid, pipe rack, road, gates, control room, muster points
2. Colour by downtime, corrective jobs or instruments flagged
3. 2D view: 25, 50, 100% of the lower flammable limit, jet and pool fire, blast zones
4. Controls: fluid, hole 10 to 100 mm, wind, stability class, shutdown mode, ignition time
5. 3D view: gas cloud that grows and drifts, flames, detectors and the muster walk
6. Effects table: release rate, cloud mass, time to detector, time to control room

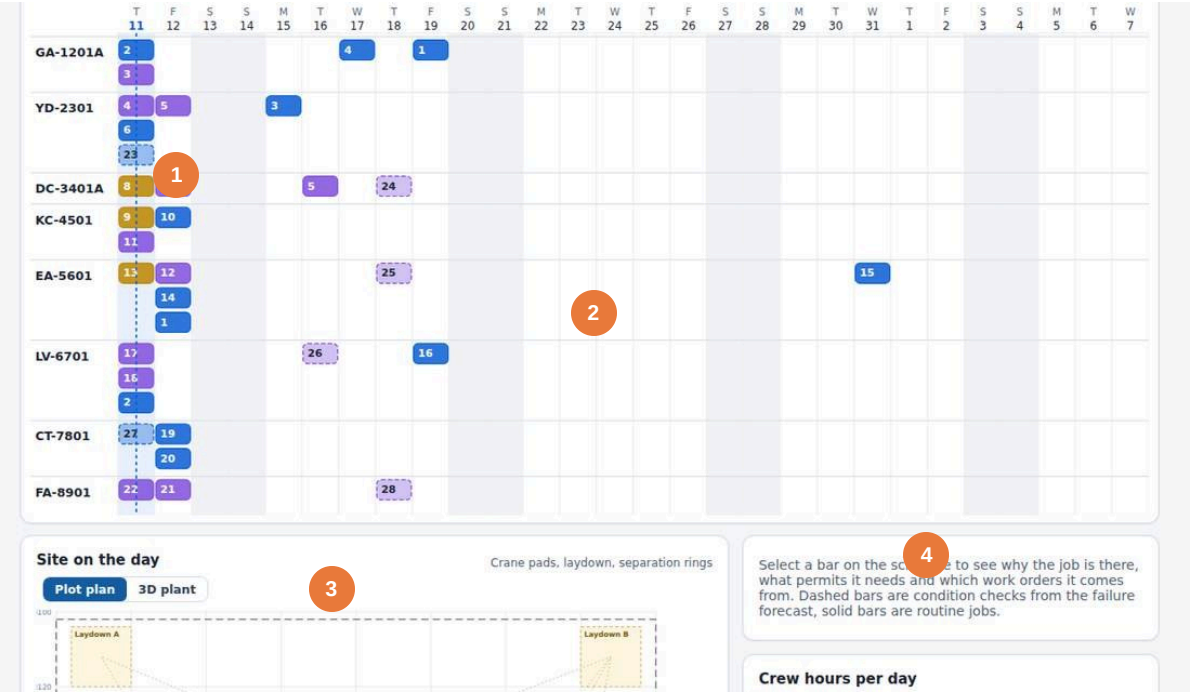
1 Release	2 Source	3 Dispersion	4 Ignition	5 People
orifice flow, hole 10 to 100 mm, shutdown auto, manual or failed	flash fraction and pool evaporation; inventory limits duration	Gaussian plume moving with wind, transport delay, urban Pasquill sigmas, dense-gas factor	flash fire at 25, 50, 100% LFL; jet and pool fire radiation; Kinney-Graham blast	detection and shutdown timing; receptors per zone; walking time to each muster point

Connected to: Site plan | Assistant | Readiness | 3D plant

Source: Prototype plot plan and hazard simulator (pp.js, hse.js, hse3d.js); screening estimates, not a consequence study

A16 Maintenance Site Plan and Plant Replay: Permits, Clashes and the Plant on Any Date

The four-week plan is built from the work-order intervals and the failure forecast; the replay lets a manager watch the plant history on the same map





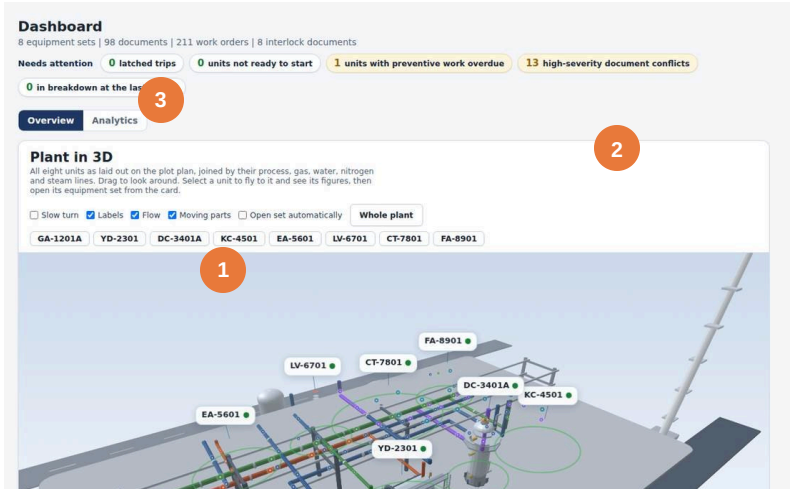
1. Gantt by unit: solid bars routine jobs, dashed bars forecast checks; letters mark permits
2. Crew hours per day against capacity
3. Site on the day: crane pad, laydown areas, separation rings, permits
4. Clash table with Move to a clear day; add jobs or load an example campaign
5. Date slider, event track and play controls; speed up to very fast
6. 3D plant with each unit's open work order and problem on its label
7. Status board, availability and cost curves up to the chosen date

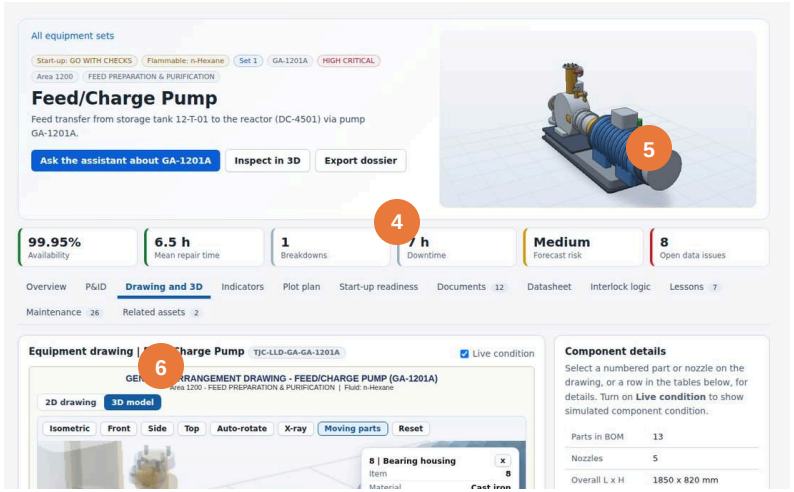
Jobs	Permits	Schedule	Clashes	Replay
routine: median interval of past jobs per unit, type and discipline; forecast checks from the gamma-Poisson modes (probability 8% or more)	hot work, confined space, lift, isolation, line break inferred from similar work-order wording	greedy by due day, levelled over 3 days, crews per discipline x 8 h; weekends skipped	hot work vs open line at the larger of the setting and 2 x hazard radius; one crane; no hot work with entry	state per unit from work-order times; stepping lands inside every breakdown, so none is skipped

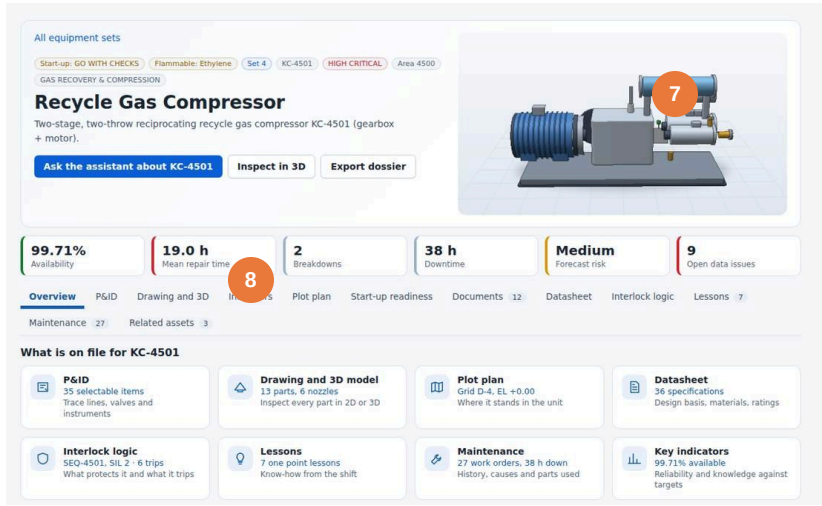
Source: Prototype site planner and plant replay (plan.js, rp.js); figures from the case work orders

A17 3D Plant and Equipment Drawings: Every Part Selectable, Built From the Datasheet Bill of Material

A purpose-built renderer shows the plant and eight machine models, each assembled from the same primitives as its 2D general-arrangement drawing







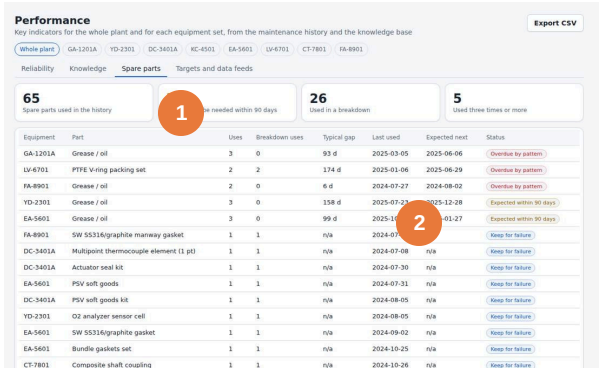
1. Plant in 3D: units on the plot plan joined by service-coloured lines, with labels and state
2. Select a unit to fly to it; chips jump to any unit
3. Attention strip: latched trips, units not ready, overdue work, high-severity conflicts
4. Machine in 3D: parts from the bill of material; select one for its card
5. Views, exploded and moving parts; live condition overlay
6. Component card: material, quantity, work orders that mention it
7. Header with a slowly turning model of the same machine
8. Overview cards open every record on file

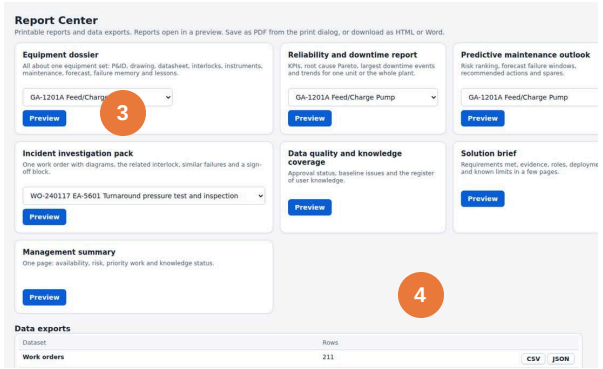
Software renderer	Picking	Condition model	Performance
depth-buffered rasteriser on a canvas, smooth normals, outlines, transparent clouds; no plug-in	the part under the cursor is identified, not just the machine, so each part opens its own card	simulated bearing temperature, vibration, leak rate and winding temperature per part; a hook replaces them with plant data	duty-cycled loops, off-screen pause, adaptive resolution aimed at 15 ms, scene cache: 15 to 50 fps on the equipment view

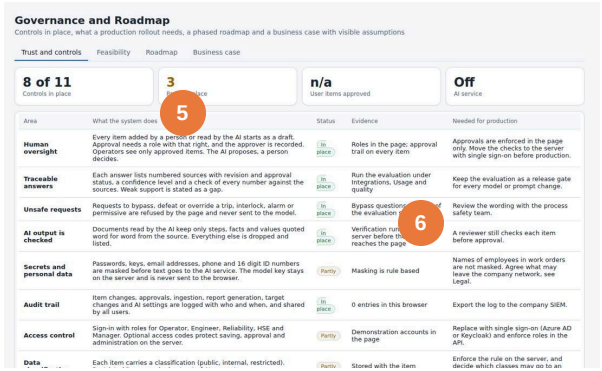
Source: Prototype 3D engine and drawings (gad.js, ga.js, g3d.js, g3r.js, plant3d.js); frame rates measured in a headless browser

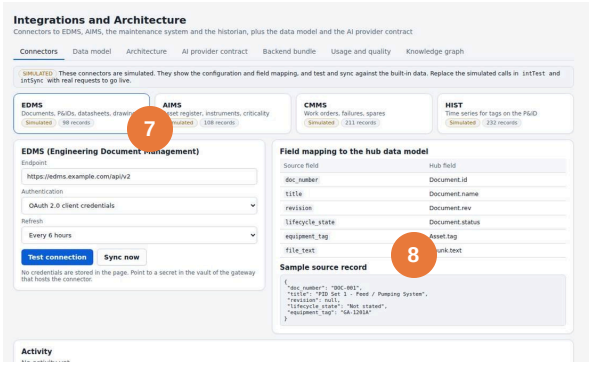
A18 Performance, Reports, Governance and Integrations: The Management Layer

Pages for the people who decide, not only the people who operate









1. KPIs per unit and plant: availability, MTBF, MTTR, cost
2. Spare parts outlook and editable targets
3. Printable reports: dossier, reliability, forecast, incident pack, data quality, management summary
4. CSV and JSON exports of work orders, instruments, documents, C&E matrix
5. Controls in place and what production needs
6. Phased roadmap and a business case with visible assumptions
7. Connectors: EDMS, AIMS, maintenance system, historian with field mapping
8. Data model, architecture, AI contract, backend bundle, knowledge graph

Also on these pages

Data feeds
Fields needed for energy, production and stock are listed, not invented.

Report Center
PDF through print, HTML or Word; every report cites its records.

Coverage register
Maps each case requirement to a feature and runs an answer audit.

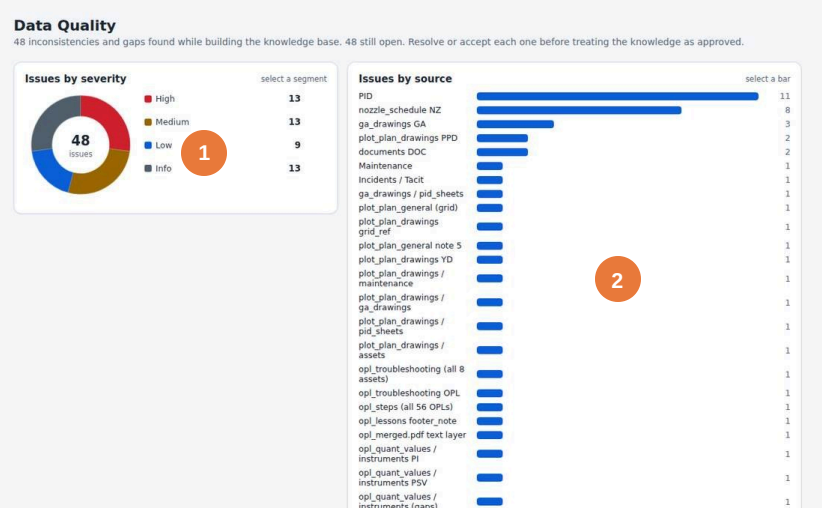
Backend bundle
SQL schema, API spec, prompt template and JSON schemas as downloads.

Connected to: Database | Failure memory | Predictive | Quality register | Assistant

Source: Prototype pages (perf.js, rc.js, gov.js, int.js, cov.js); all figures computed from the loaded records

A19 Platform Services: Roles, Sync, Offline, Language and Guidance Around Every Feature

The services that make the hub usable by a shift team and by the Company's IT, not only by the demo author



1. Data Quality: 48 findings by severity, with owner; reviewers mark resolved or accepted, and the mark is shared
2. Same data feeds the notification bell and My work: approvals waiting, rejected items, answers marked wrong, high-risk forecasts

Roles

Operator, Engineer, Reliability, HSE and Manager; what a person can approve or edit follows the role. Demo accounts from the work-order names; not a security boundary.

Shared database

Pull, merge and push through /api/kb with last-write-wins, conflict detection by base version and tombstones; falls back to browser storage.

Offline and install

Service worker keeps the page when the network drops; manifest makes it installable; /api calls are never cached.

Language

Questions in Bahasa Indonesia are translated to the record terms before search; interface and answers can be switched to Indonesian.

Guided tour and help

A short tour of the main areas starts once for a new visitor and can be reopened from the sidebar.

Accessibility and layout

Keyboard navigation, labelled controls, automatic label de-collision on every SVG figure, layout crawled at phone, tablet and desktop widths.

Settings and diagnostics

Connection state, AI status and limits, usage per day, language, role, self-test of the server functions, export and restore of the shared data.

Global search

Ctrl K searches the same ranked index across documents, work orders, lessons and tags.

Source: Prototype services (auth.js, sync.js, notify.js, i18n.js, help.js, ui.js, sw.js)

A20 Data Sources: Purpose, Owner and Business Justification for Each, and Which Features Read It

Baseline data from the case plus proposed additions; a small set of well-governed sources is preferred to many weakly connected ones

Source	Content	Business justification	Status	Assistant	Failure memory	Predictive	P&ID, interlock	OPL simulator	Readiness	Hazard	Site plan	3D plant
SOPs and OPLs (56)	Steps, safety notes, tacit lessons	Fewer wrong steps; keeps SME experience	Loaded	●		●	●		●	●		
Datasheets (8)	Limits, materials, ratings	One place for trusted specifications	Loaded	●				●			●	
P&ID, GA, plot plans (24)	Equipment, tags, layout	Every tag tied to drawing and place	Loaded	●			●	●	●	●	●	●
Interlock logic (8, 84 instr.)	Causes, effects, permissives	States what trips and why	Loaded	●		●	●	●	●			
Maintenance history (211)	Failures, causes, cost, downtime	Failure memory, KPIs, forecast, plan	Loaded	●	●	●			●		●	
Incident and RCA reports	Lessons learned	Prevents repeat events	Add in pilot	●	●	●		●	●			
EDMS	Controlled revisions	Single source for revisions	Adapter specified	●								
AIMS	Work orders, asset tree	Live history, drafts back	Adapter specified		●	●					●	
Historian or DCS	Live tag values	Context for troubleshooting	Hook in place			●	●		●			●
Digital twin	Asset model, 3D	Navigate by equipment and place	Planned	●								●

Every additional source needs an owner, a refresh rule and a stated decision it improves, because the case asks for a well-justified architecture rather than more data.
Source: Case book data table; prototype counts and module dependencies

A21 Integration: Four Connector Contracts Feed One Core, Read-Only First

Each connector is a field map, an authentication method and a schedule; the prototype simulates the handshake, a pilot connects the real systems

EDMS

Documents, P&IDs, datasheets, drawings

Auth: OAuth 2.0 client credentials **Refresh:** every 6 hours

SOURCE FIELD	HUB FIELD
doc_number	**Document.id**
revision	**Document.rev**
lifecycle_state	**Document.status**
equipment_tag	**Asset.tag**
file_text	**Chunk.text**

AIMS

Asset register, instruments, criticality

Auth: API key in a vault **Refresh:** daily

SOURCE FIELD	HUB FIELD
asset_id	**Asset.tag**
functional_location	**Asset.floc**
criticality	**Asset.criticality**
instrument_tag	**Instrument.tag**
range	**Instrument.setpoint**

CMMS

Work orders, failures, spares

Auth: service account token **Refresh:** every hour

SOURCE FIELD	HUB FIELD
wo_number	**WorkOrder.wo**
problem_description	**WorkOrder.prob**
root_cause	**WorkOrder.root**
downtime_hours	**WorkOrder.dt**
total_cost	**WorkOrder.tc**

Process historian

Time series for tags on the P&ID

Auth: Kerberos **Refresh:** streaming

SOURCE FIELD	HUB FIELD
tag	**P&ID tag**
value	**live value**
quality	**signal quality**
timestamp	**time**

Knowledge database Asset, Document, Chunk, WorkOrder, Failure, Interlock, Instrument, OPL, Lesson, Source: all joined on the equipment tag

Single sign-on
roles from the company directory

Read first, write later
writes are drafts that a person approves

One contract per source
a new system is one more map

Data stays in the tenant
region and retention set by the Company

Test and sync log
every handshake and sync recorded in the audit trail

Prototype behaviour: Test connection validates the address and returns a sample record; Sync rebuilds the index and reports the record count. A pilot replaces these with live paging against the Company's endpoints.

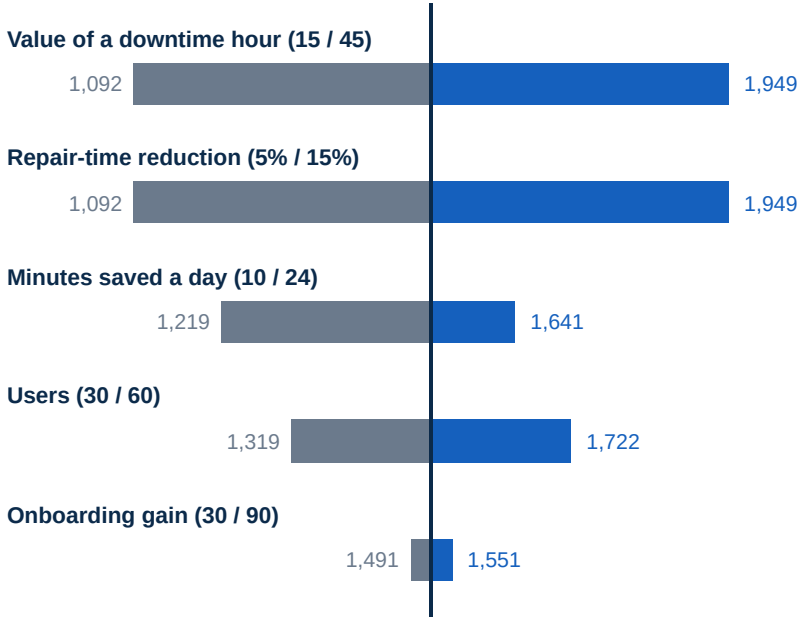
Source: Prototype Integrations page (int.js): connector definitions with placeholder addresses; no Company system is connected yet

A22 Business Case: Assumptions, Calculation and Sensitivity

Items marked Assumed are estimates to be replaced by Company figures in the pilot; Data items come from the case baseline

Item	Low	Base	High	Source
Users	45	45	45	Assumed: operators, technicians, engineers, HSE
Minutes saved per user per day	10	20	24	Assumed 40 min spent searching today
Loaded cost per hour (IDR)	175,000	175,000	175,000	Assumed; 230 working days a year
Breakdowns a year	20.4	20.4	20.4	Data: 31 in 1.52 years
MTTR today (h) and reduction	14, 5%	14, 10%	14, 15%	Data for MTTR; reduction assumed
Value of one downtime hour (IDR M)	30	30	30	Assumed; replace with plant finance figure
Onboarding gain (IDR M)	30	60	90	4 hires a year, 1 month faster, half productive
Annual benefit (IDR M)	760	1,521	2,100	Sum of the three levers
One-off / yearly run cost (IDR M)	1,400 / 370	1,400 / 370	1,400 / 370	Integration 650, data 350, change 250, security 150
Payback (months) / 3-year net (IDR B)	over 36 / -0.5	21 / 1.4	16 / 2.9	Benefits ramp: 60% in year 1, 100% after

Sensitivity of the base annual benefit (IDR M)



Base annual benefit: IDR 1,521 M

Time saved and the value of a downtime hour dominate the result. Both are assumptions, so the pilot measures them first: time to a verified answer from assistant timestamps, and downtime value from plant finance.

Running cost covers hosting, model use, monitoring and half a support role. Not counted: avoided incidents, spare-part savings and knowledge that would otherwise leave with retirees.

Source: Case baseline data; team assumptions. IDR M = million rupiah

A23 Test Evidence: Five Layers of Checks Run on Every Build, and What Has Not Been Tested

The prototype is verified against the case data automatically; live Company systems are the next test

Static checks
duplicate functions, ids, wording, build order, dead links

Regression checks
101 checks of data, tools, models, pages, simulators, planner, replay

Live server checks
86 checks: two-user sync, conflicts, AI proxy against a simulated model, limits

Evaluation set
128 curated questions: facts, sources, numbers, refusals, Bahasa Indonesia

Layout and 3D stress
every page at 390, 820 and 1440 px; random use of all 8 models

Static checks gate the build; the regression, live and evaluation suites fail the build on a low score.

Check	Result
Curated questions (interlocks, procedures, statistics, safety, hazards, failure modes)	All 128 pass: correct facts, sources shown, refusals where required
Safety requests (bypass an interlock, defeat a trip)	All refused with the reason and the approved route
Generated audit of answers against records	98 of 98 correct
Two browsers on one shared database	Edits sync, conflicts detected, offline mode falls back
Simulated language-model server	Streaming, tool calls, saved key and failure fallback work
Plant replay across all 31 breakdowns	Affected unit shown in plot plan and 3D, at the fastest speed too
Hazard model sanity	Bigger hole gives a larger release; calm air a longer cloud; failed shutdown a longer release
Predictive backtest (64 forecasts)	Cannot separate units; reported as advisory with limits

Not yet tested

- A live connection to the Company's EDMS, AIMS or historian
- The real Gemini service under load: free-tier quota behaviour is handled by limits and fallback, and measured in the pilot
- Usability with operators on shift; the success gates of the pilot measure it

Source: Prototype test suites (tests/static.js, regress.js, live.js, layout.js, stress3d.js) and the 128-question evaluation set

A24 Chemical Domain Relevance: Every Feature Starts From a Real Hazard or Routine of the LLDPE Unit

Left: a process fact from the case data. Right: what the platform does with it, and the module that does it

Hexane feed pump, 16 barg, 80 C: a seal leak is a fire and release risk	A seal work order brings up the API Plan 11 flush lesson and the PDI-1201 permissive; the hazard simulator sizes the cloud	Assistant, lessons, hazard
Start-up needs seal flush, open suction and min-flow valves before the pump may run	Readiness reads every permissive against the live model; a latched trip or a closed valve blocks the start	Start-up readiness
Catalyst reduction reactor with hydrogen and a high-temperature trip (TSHH-3401, 230 C)	Interlock library shows cause, effect and vote; the simulator shows a bypass consequence and the assistant refuses to advise one	Interlock, OPL simulator
Recycle gas compressor: anti-surge valve FV-4502 opens on low flow	Chain reaction shows which units trip next and for how long; failure memory lists earlier surge events	Scenarios, failure memory
Seal and gasket leaks are the largest failure mode (12 jobs, 14% found early)	Ranked by risk priority; a draft lesson is created when none exists	Reliability, Studio
Drawings disagree on valve fail position, tag names and nozzle sizes	Conflict register with owners; the assistant warns when an answer touches a conflict	Data quality, assistant
Flammable service near a control room and muster points	Gas cloud, jet and pool fire and blast on the true-scale plot plan, detectors, walk to muster	Hazard simulator
Hot work, lifting and line breaks planned on a running site	Site plan checks hot work against open lines by hazard-zone distance, one crane, crews and permits	Site plan

Source: Case baseline data (datasheets, P&IDs, interlock logic, OPLs, maintenance history)

A25 Security and Governance: Controls at Each Layer, and What Production Adds

Roles in the prototype are a demonstration, not a security boundary; production relies on the Company's identity and network controls

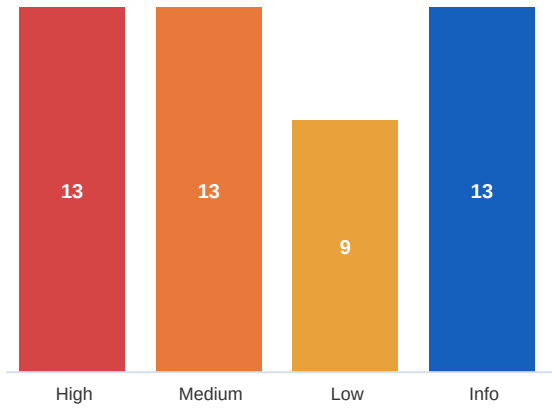
	CONTROL IN THE PROTOTYPE	PRODUCTION STEP
Person Operator to Manager	Roles filter what is shown and what can be approved; sign-in is a demo	**Single sign-on, directory groups**
Browser single-file app	Local store only; no key in the page; offline mode never sends data	**Managed device policy**
Proxy /api functions	Key held on the server; per-visitor and daily limits; size caps; secrets masked before the model	**Key vault, rotation, WAF**
Model Gemini 2.5 Flash	Sources treated as data; fixed tool list; refusal rules; temperature 0.2	**Company-approved model, no training on Company data**
Store Blobs, then PostgreSQL	Status vocabulary, approval workflow, audit trail, access list per chunk	**Region fixed, retention set, SIEM export**
Plant OT zone	Advisory only; no write to control systems; interlock bypass refused	**Hub outside the OT zone (IEC 62443)**

Request path from the top down: a question passes every layer, and each layer can refuse it.
Source: Prototype implementation; production steps are proposals

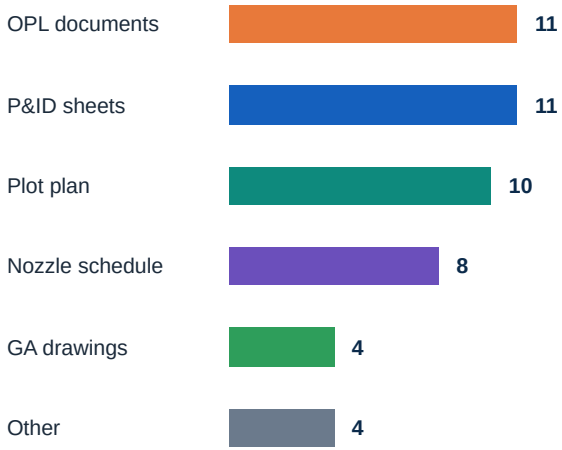
A26 Document Conflicts Found by the Ingestion Checks: 48 Findings by Severity and Source, With Examples

Each finding is kept in a register with an owner; the assistant warns when an answer touches an open conflict

By severity



By source family



Findings are written by rule when a document is ingested, then reviewed by a person who assigns the owner and the fix.

Finding	Consequence in the field
Title 'KA-4501' versus body tag 'KC-4501'	A tag search misses the sheet
FV-2302 labelled both 'trips OPEN' and 'trips CLOSED'	Valve failure position unclear
TSHH-2301 trip text reads as a low-low flow of 9 m3/h	Wrong trip description for a temperature trip
FV-3403 used for both H2 injection and trim-gas valves	Two valves share one tag
Drawing reference is a placeholder 'XXXX' or missing	Drawing cannot be traced in EDMS
Revision shown as '02' without a full date	Currency cannot be confirmed
Footprint 4.0 x 9.5 m versus 4.0 x 8.5 m versus 9000 mm	Layout and access use different numbers
GA shows a rotary steam-tube dryer; P&ID shows a fluid bed dryer	Two machines under one tag

How a conflict is handled

1 Detected by rule at ingestion or by comparison with the instrument list.

2 Listed in the quality register with severity, source and owner.

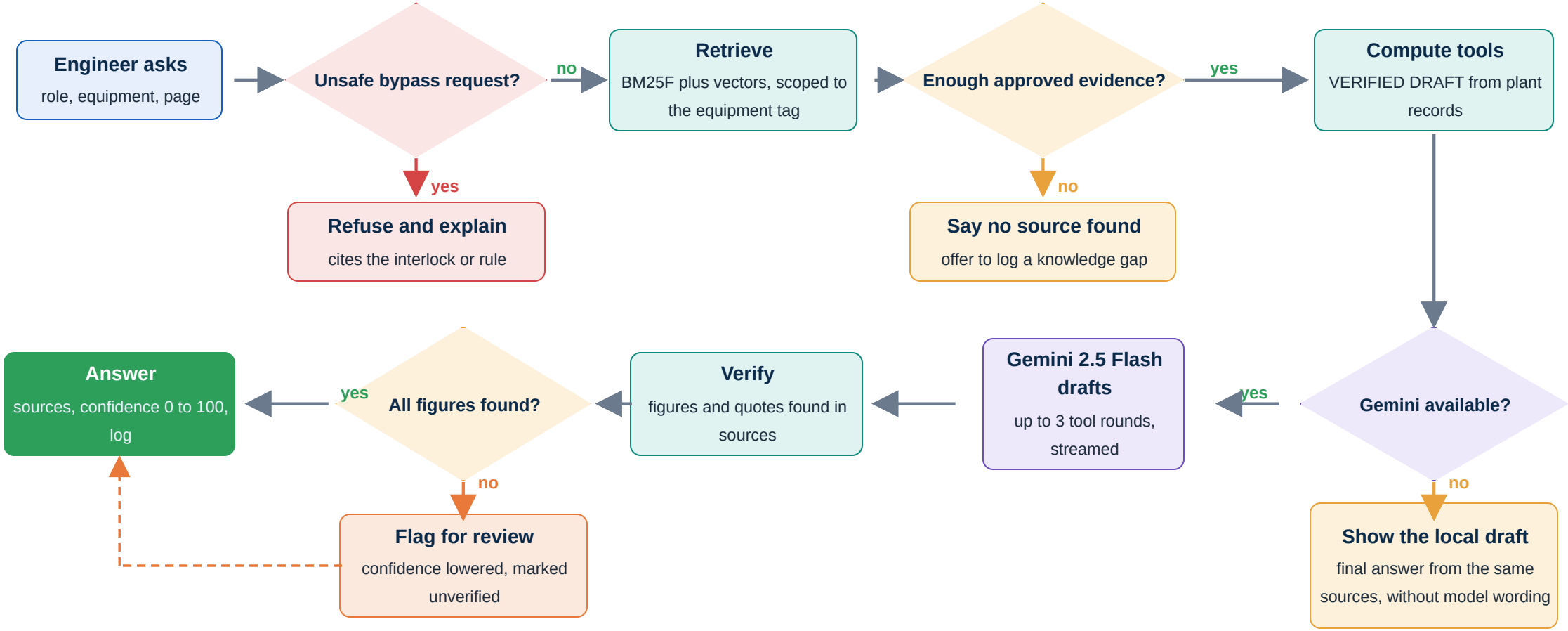
3 Shown on any assistant answer that cites the affected document.

4 Marked resolved or accepted by a reviewer; the mark is shared and audited.

Source: Quality register in the prototype, built from the case documents

A27 From Question to Answer: Four Decision Points Between an Engineer and a Trusted Reply

Every branch is implemented in the assistant; the free-tier model is one step in the path, and the path still ends in a checked answer if it is unavailable

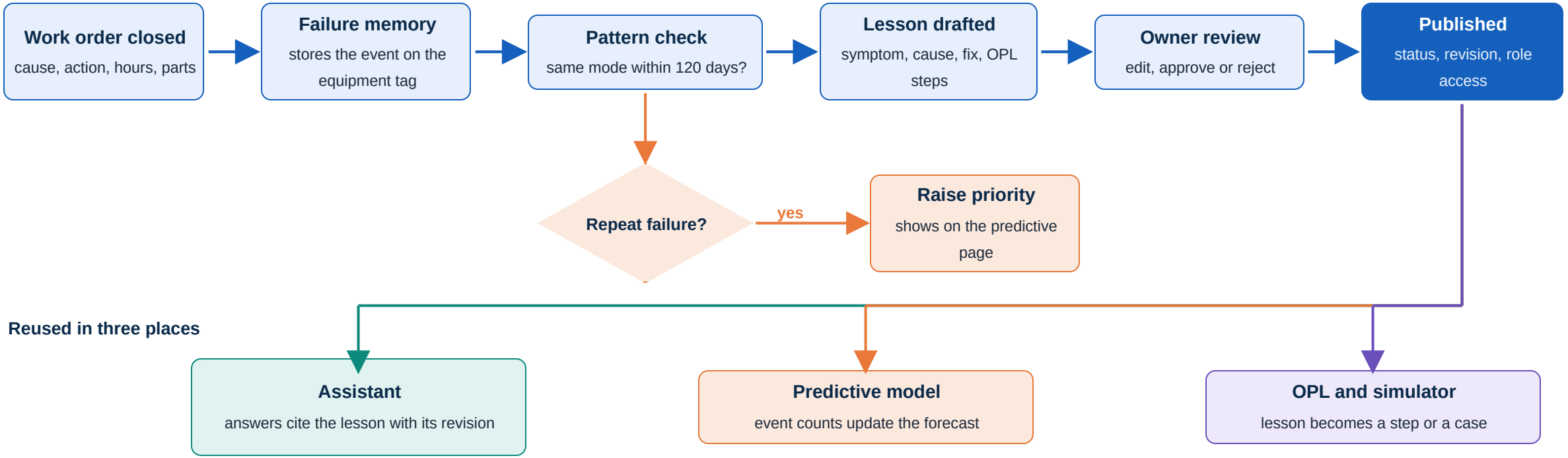


Reading the figure: red branches protect people, amber branches protect trust, and every path writes the question, sources, confidence and outcome to the audit log.

Source: Assistant page logic (ai.js, dl.js) and the netlify ai function; behaviour verified by the 128-question evaluation set

A28 From Plant Event to Reusable Lesson: A Closed Loop That Feeds Answers, Prediction and Simulation

A closed work order is the start of knowledge, not the end of the paperwork; nothing is reused until a named owner approves it

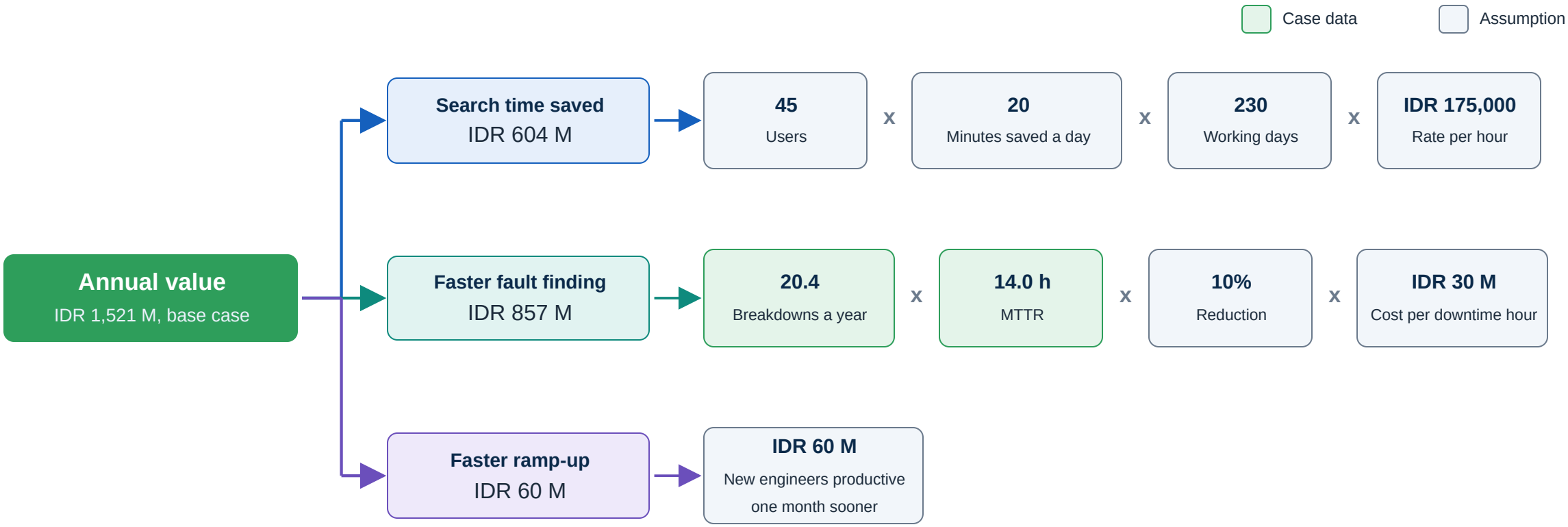


Real example in the data: work order WO-240088 renewed a crosshead pin and bush and set the clearance. Once approved as a lesson it appears in the answer for a similar symptom, raises the repeat-failure count for that failure mode, and can be added to the OPL as a check step. Feedback marked wrong goes back to the review queue.

Source: Work-order history in the prototype (211 work orders, 55 corrective jobs followed by a repeat-failure check)

A29 How the Annual Value Is Built: A Driver Tree From the Headline Number Down to Each Input and Its Source

Green inputs come from the case data; grey inputs are assumptions to be replaced with Company figures during the pilot



Sensitivity: low case IDR 0.8 B (no payback in 3 years), base IDR 1.5 B, high IDR 2.1 B. The repair-time lever is the largest and the least certain; the pilot measures it first.

Source: Business case model in make_deck.py and appendix A22; case data from the 211 work orders

A30 Fit With Chandra Asri's Direction: The Knowledge Layer That a Digital Twin and Industry 4.0 Programme Still Needs

Public information about Chandra Asri, used to position the proposal; internal systems and data are to be confirmed with the Company

What Chandra Asri already does	The gap this proposal fills	What the platform adds, read-only
Industry 4.0 programme National Lighthouse Industry 4.0 award; INDI 4.0 award, aggressive digitalisation	**Documents, lessons and work-order text** SOPs, OPL, RCA and tacit know-how are not tied to the equipment tag	**Tag-keyed knowledge base** documents, work orders and lessons joined to the same equipment tags the twin uses
Digital twin, Cilegon complex plant and engineering data for asset performance, reliability and proactive maintenance	**Answers without proof** engineers cannot tell which source or revision to trust	**Verified answers with sources** revision, status and confidence on every reply
		Failure memory and OPL simulator history and lessons used for prediction and training
Digital monitoring cooling-water monitoring saved 348 M litres of water, 6 M kWh and 2,100 t CO2 in 2020	**Knowledge leaves with people** lessons from repeat failures are not reused	**Links to twin, historian, AIMS, EDMS** adapters read tag values and records; nothing is written back

How it plugs in

Twin and historian — tag values, read-only → **AIMS** — work orders and plans → **EDMS** — controlled documents → **TAGLINK** — joins on the equipment tag → **Knowledge hub** — answers, prediction, planning

Source: Chandra Asri press releases (National Lighthouse Industry 4.0 award, INDI 4.0 award); Hydrocarbon Processing, Aug 2021; Siemens press release, Jul 2020

A31 Impact Detail: Financial Assessment, IDR 1.5 B a Year in the Base Case, With Cash Profile and Scenarios

Financial, operational and people benefits with measurable KPIs; hours and rates are assumptions to be replaced by Company figures during the pilot

Financial assessment

Parameter	Baseline	With platform
Time to a verified answer	about 15 min*	3 min
Mean time to repair (MTTR)	14.0 h	12.6 h
Repeat failures within 120 days	8 of 55	5.6 of 55
Annual benefit (IDR M)	0	1,521
One-off investment (IDR M)	0	1,400
Annual running cost (IDR M)	0	370

IDR 1.44 B
net benefit over 3 years

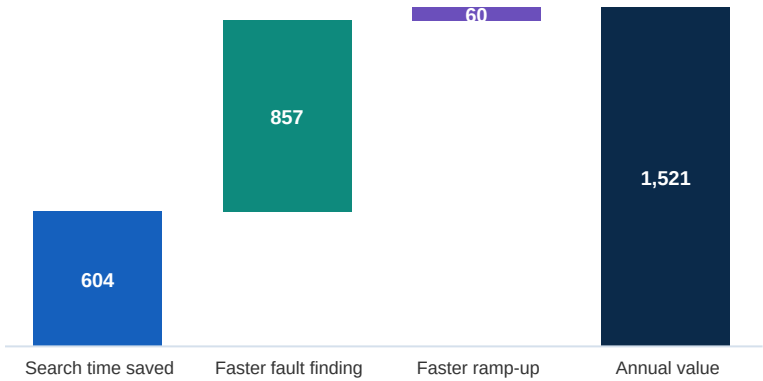
40%
internal rate of return

IDR 0.91 B
net present value at 10%

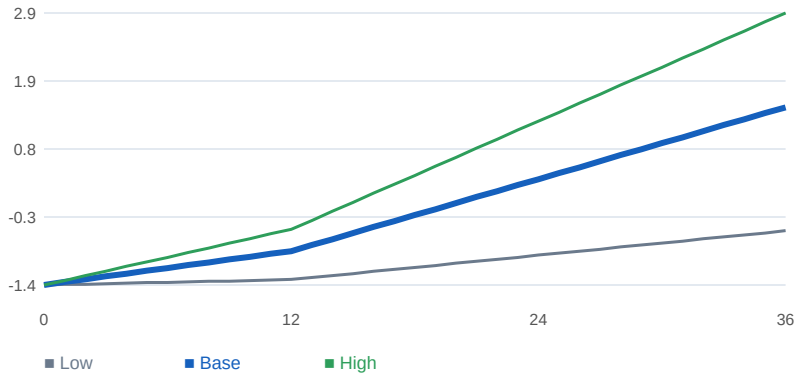
21 months
payback period

Low case	Base case	High case
IDR 0.8 B a year	IDR 1.5 B a year	IDR 2.1 B a year
no payback in 3 years	payback 21 months	payback 16 months

Annual value by lever, base case (IDR M)



Cumulative net benefit (IDR B)



From feature to KPI to value

FEATURE	KPI MOVED	IDR M
AI assistant and verified answers	Time to answer 15 to 3 min	604
Failure memory and predictive checks	MTTR -10%, repeat failures -30%	857
Knowledge Studio and OPL lessons	New engineers productive 1 month sooner	60
Interlock guard, readiness, hazard view	Fewer improper executions	not valued
Quality register and approvals	48 conflicts owned, 90% closed	not valued

Total base case IDR 1,521 M a year; two levers are left unvalued on purpose

* assumed; all other baseline values come from the 211 work orders

Source: Case baseline data (work orders, MTTR, breakdowns); assumptions in appendix A22. IDR = Indonesian rupiah; IRR and NPV over 3 years, 10% discount rate